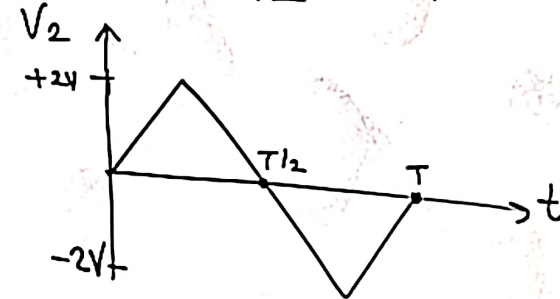
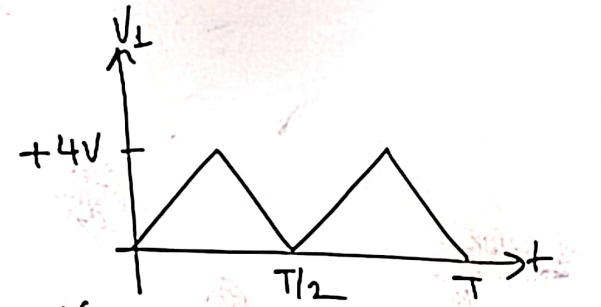
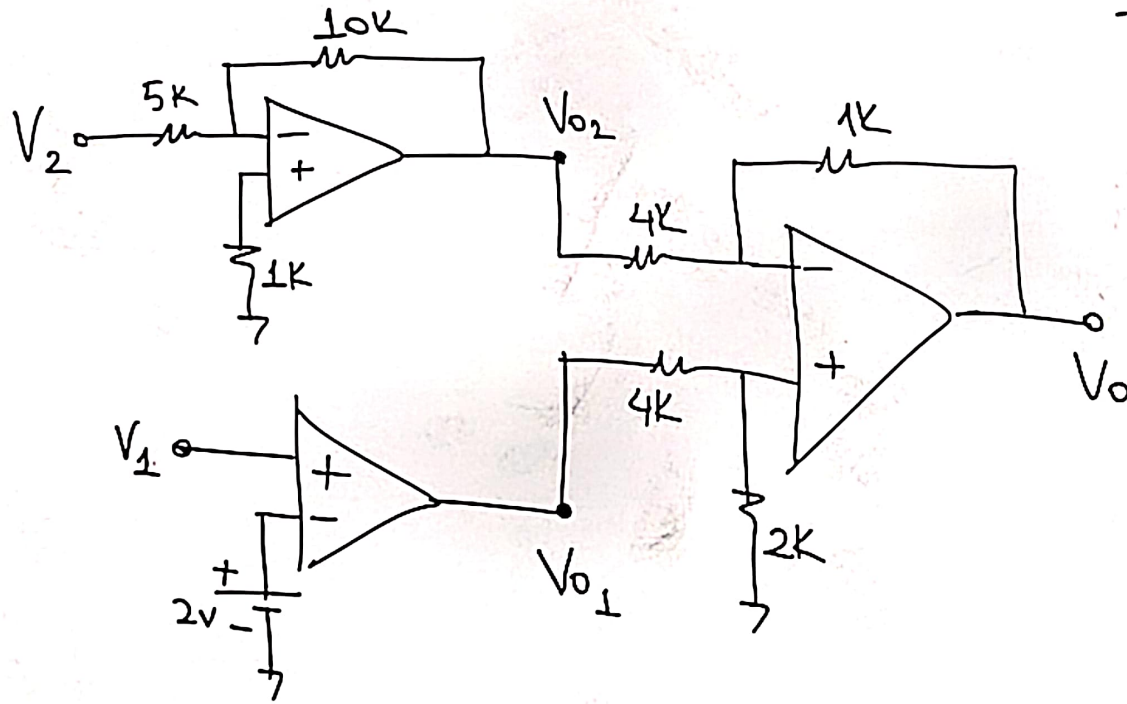


2024-Büt.

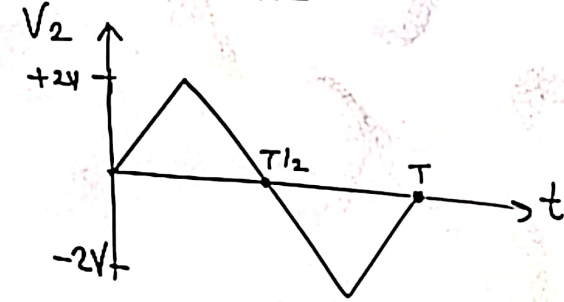
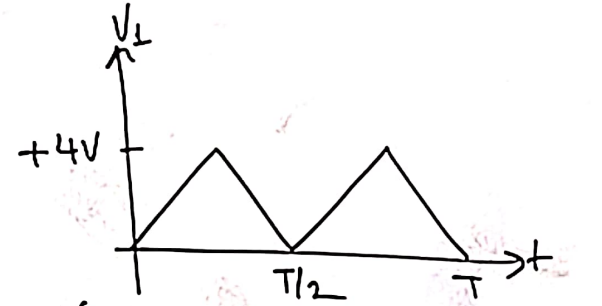
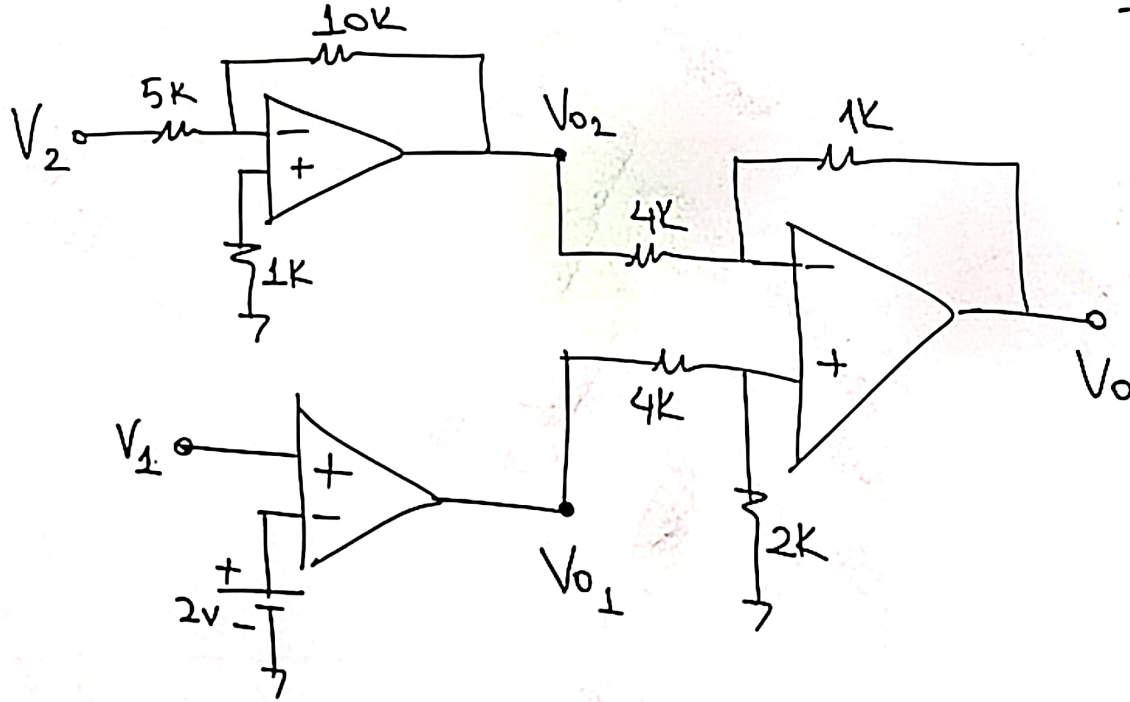


$$\pm V_{CC} = \pm 12V$$

$$T = 16ms$$

V_{01}, V_{02}, V_0 olarak çiz.

2024-Büt.

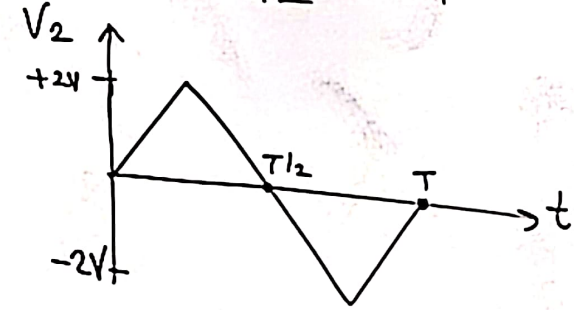
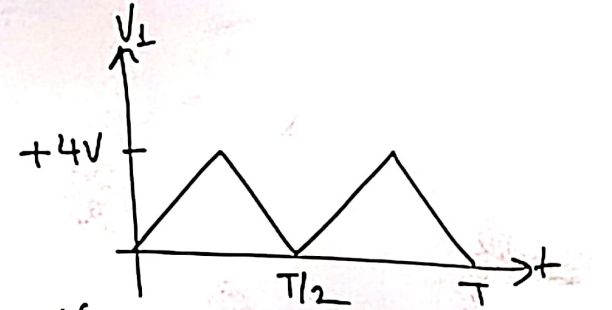
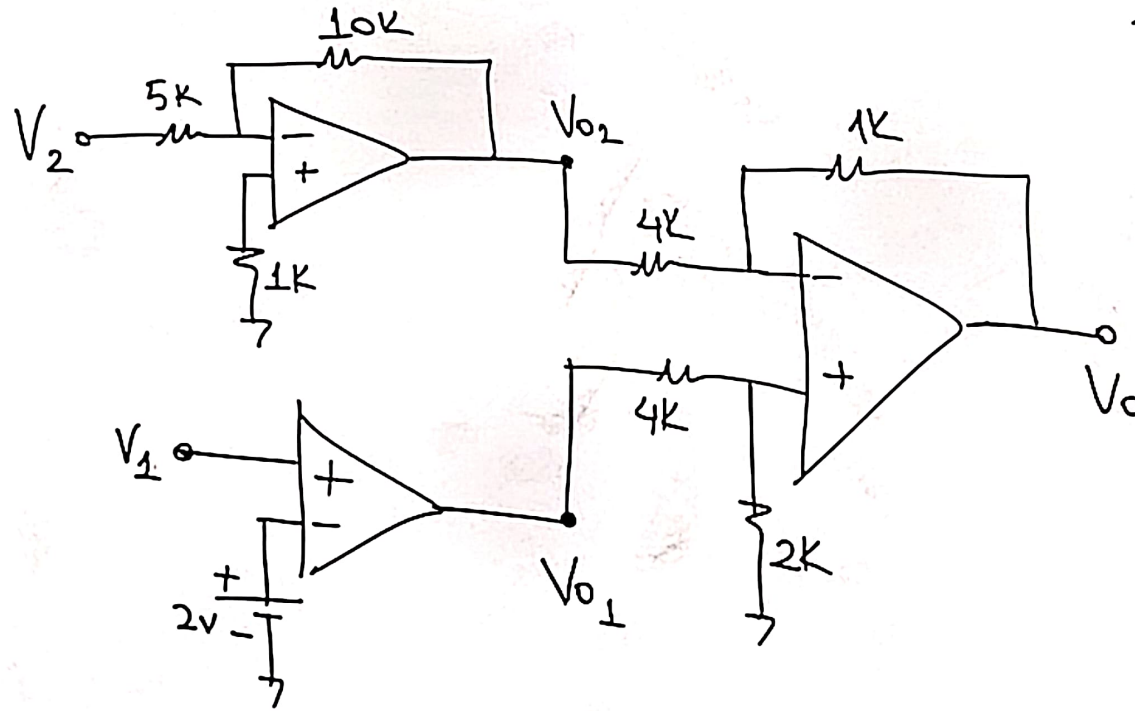


$$\pm V_{CC} = \pm 12V$$

$$T = 16ms$$

V_{01}, V_{02}, V_0 'ı özetli olarak çiz.

2024-Büt.

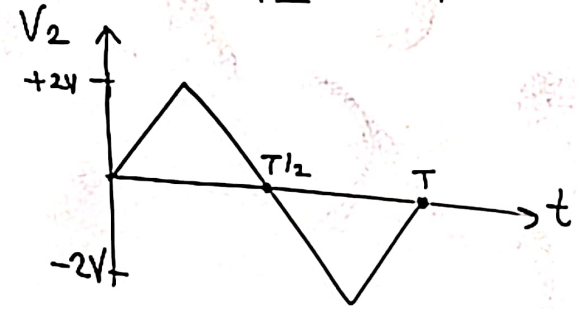
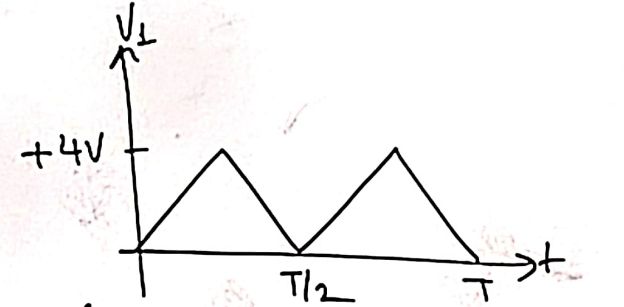
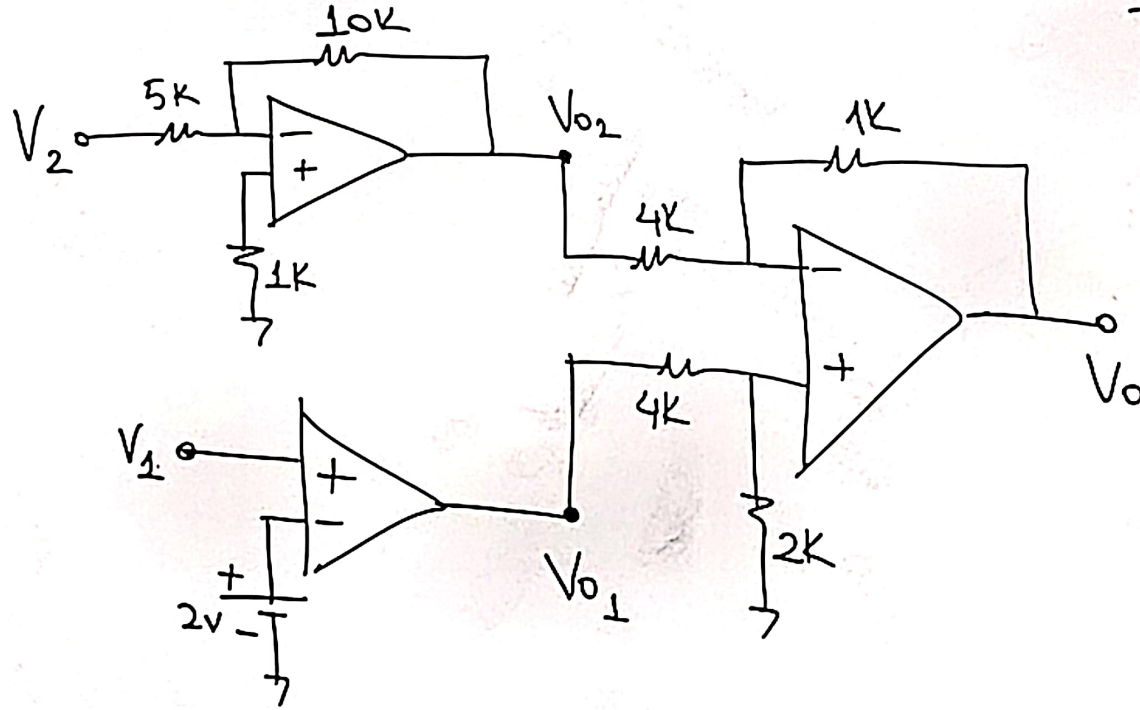


$$\pm V_{CC} = \pm 12V$$

$$T = 16ms$$

V_{O1}, V_{O2}, V_0 'ı örnekli olarak çiz.

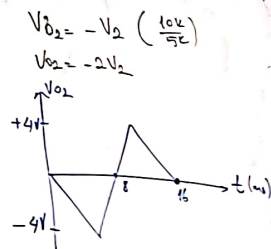
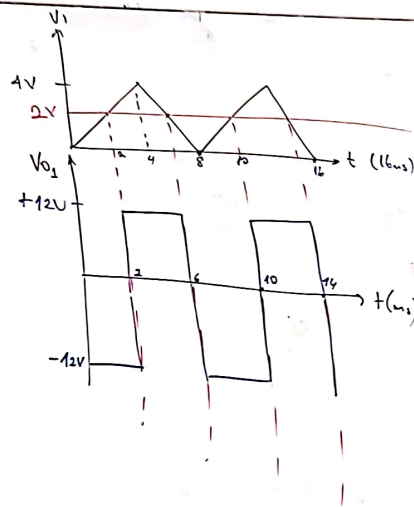
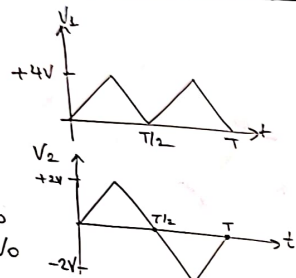
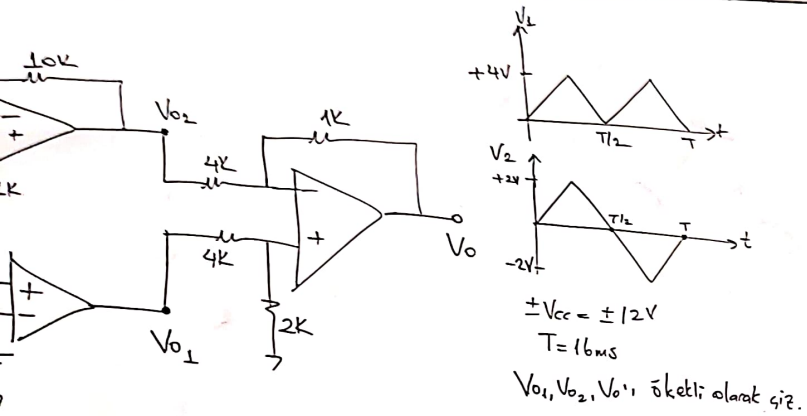
2024 - Büt.



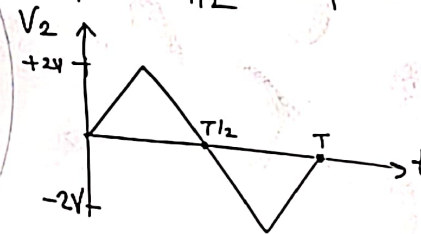
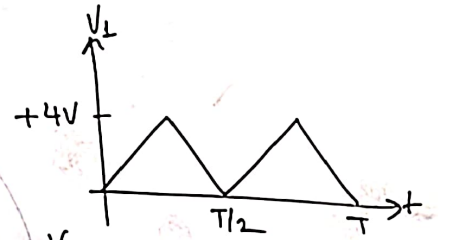
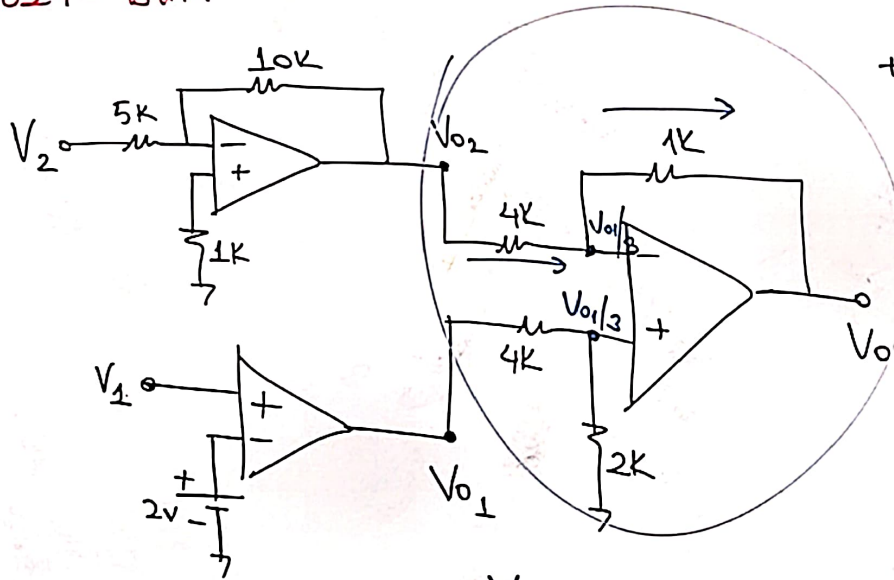
$$\pm V_{CC} = \pm 12V$$

$$T = 16ms$$

V_{01}, V_{02}, V_0 'ı örnekli olarak çiz.



2024- Büt.



$$\pm V_{cc} = \pm 12V$$

$$T = 16ms$$

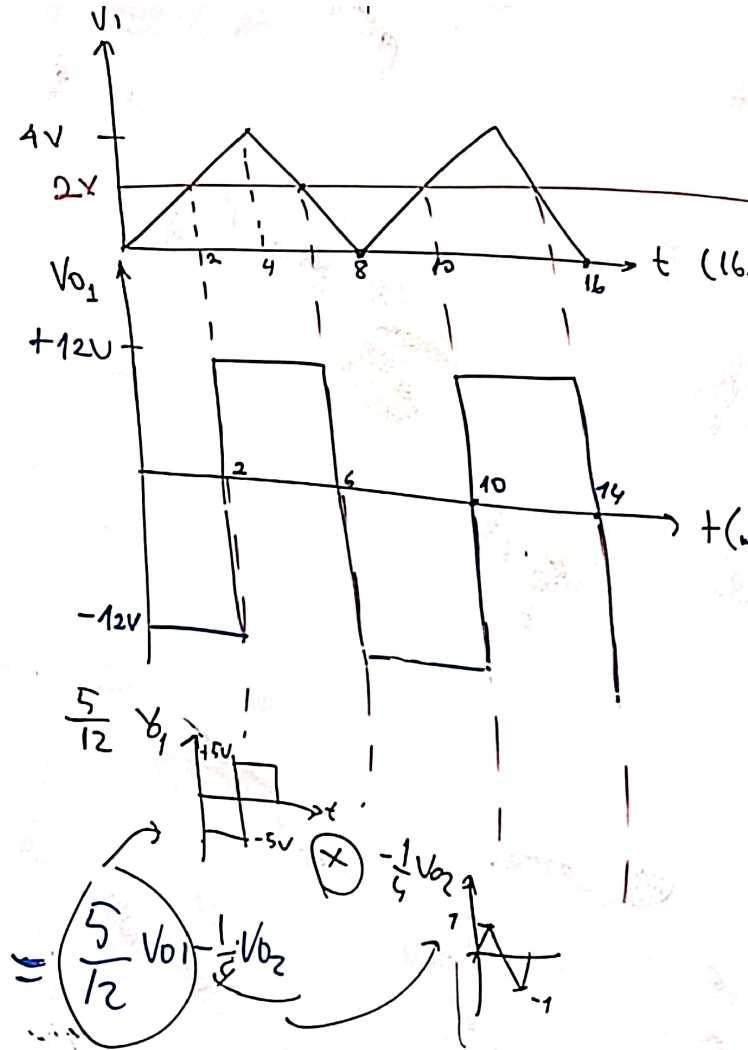
$V_{01}, V_{02}, V_{01/3}$ ötekli olarak çiz.

$$\frac{V_{01}}{6K} \cdot 2K = V_+$$

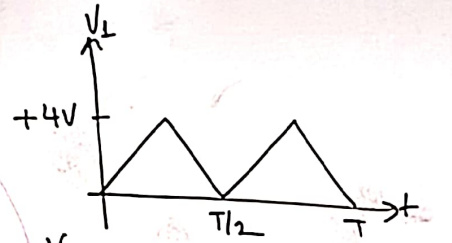
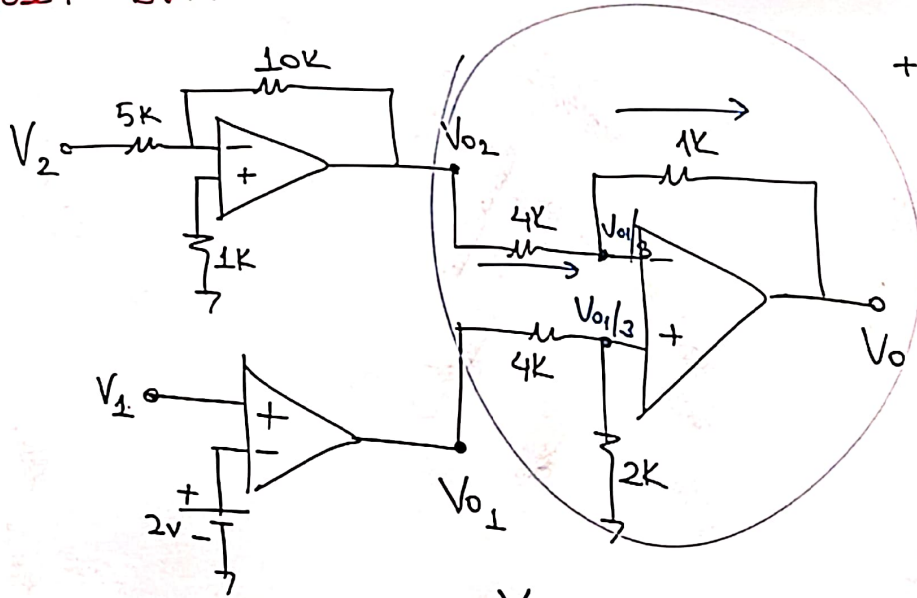
$$\frac{V_{02} - V_{01/3}}{4K} = \frac{V_{01/3} - V_0}{1K}$$

$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

$$4V_0 = \frac{5}{3}V_{01} - V_{02} \Rightarrow V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right) = \left(\frac{5}{12}V_{01} - \frac{1}{4}V_{02} \right)$$



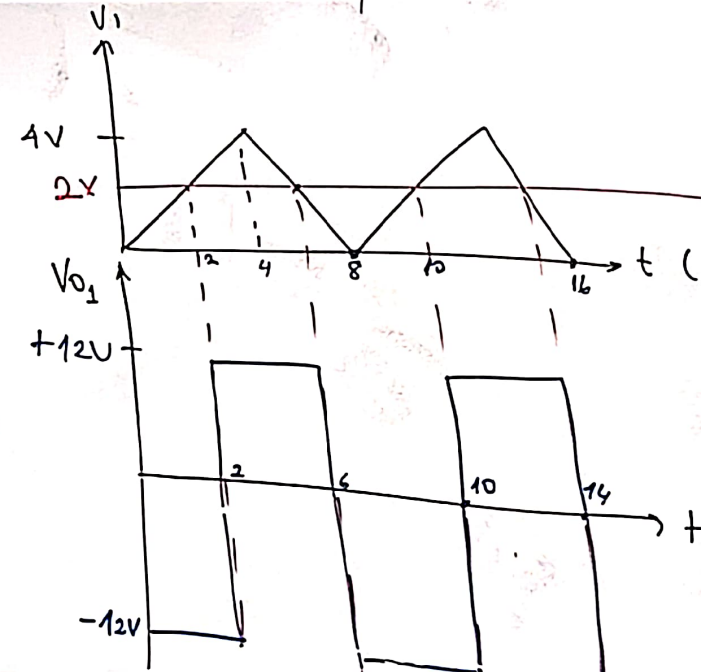
2024-Büt.



$$\pm V_{CC} = \pm 12V$$

$$T = 16ms$$

\$V_{01}, V_{02}, V_0\$'ı örnekli olarak çiz.

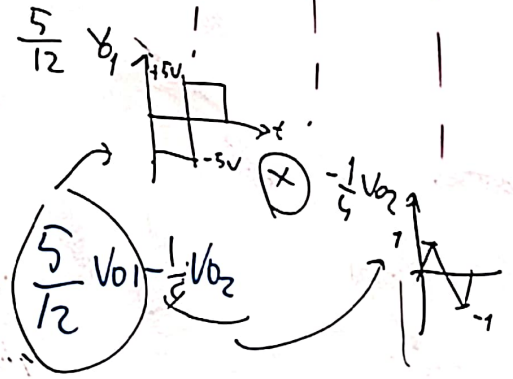


$$\frac{V_{01}}{6K} \cdot 2K = V_+$$

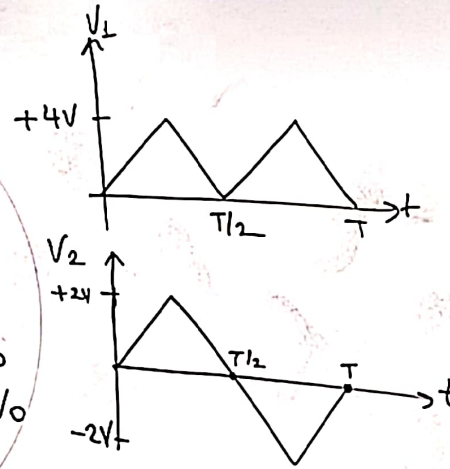
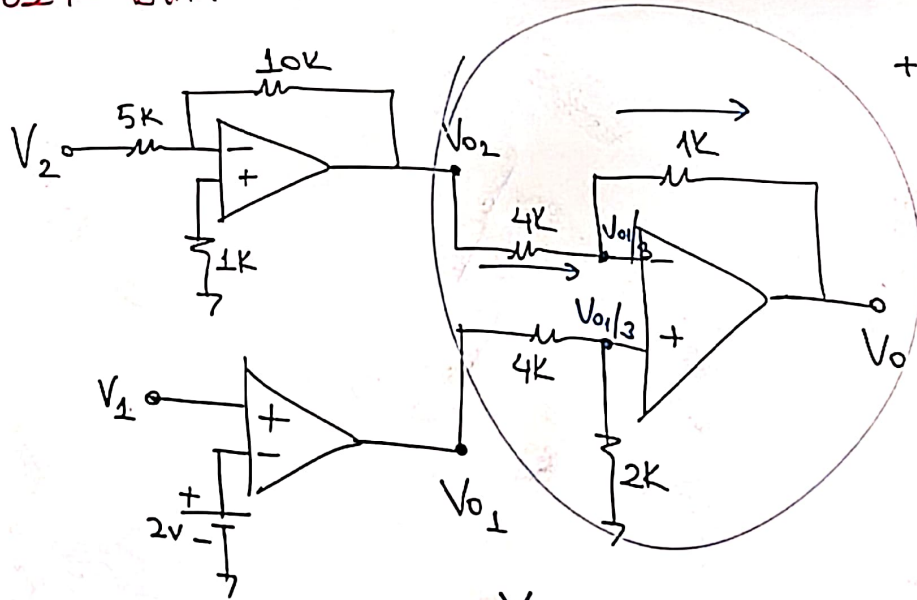
$$\frac{V_{02} - V_{01}/3}{4K} = \frac{V_{01}/3 - V_0}{1K}$$

$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

$$4V_0 = \frac{5}{3}V_{01} - V_{02} \Rightarrow V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right) = \left(\frac{5}{12}V_{01} - \frac{1}{4}V_{02} \right)$$



2024- Büt.



$$\pm V_{cc} = \pm 12V$$

$$T = 16ms$$

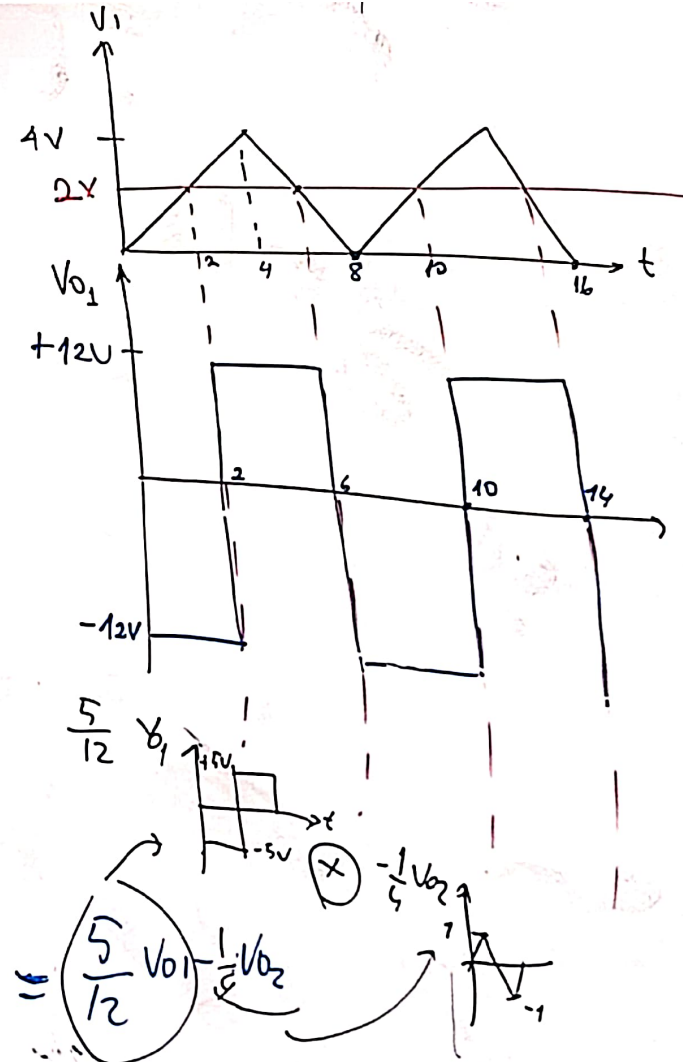
V_{o1}, V_{o2}, V_o olarak çiz.

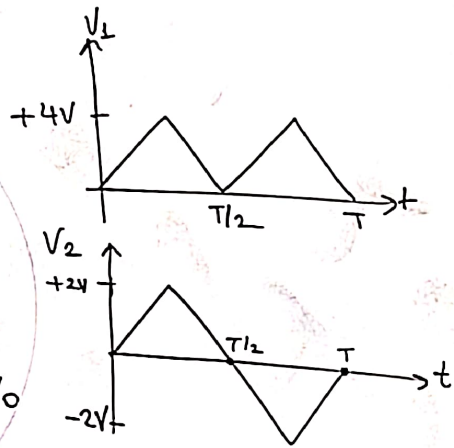
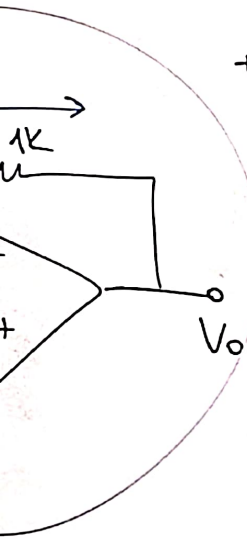
$$\frac{V_{o1}}{6K} \cdot 2K = V_+$$

$$\frac{V_{o2} - V_{o1/3}}{4K} = \frac{V_{o1/3} - V_o}{1K}$$

$$V_{o2} - \frac{V_{o1}}{3} = \frac{4}{3}V_{o1} - 4V_o$$

$$4V_o = \frac{5}{3}V_{o1} - V_{o2} \Rightarrow V_o = \frac{1}{4} \left(\frac{5}{3}V_{o1} - V_{o2} \right) = \left(\frac{5}{12}V_{o1} - \frac{1}{4}V_{o2} \right)$$





$$\pm V_{CC} = \pm 12V$$

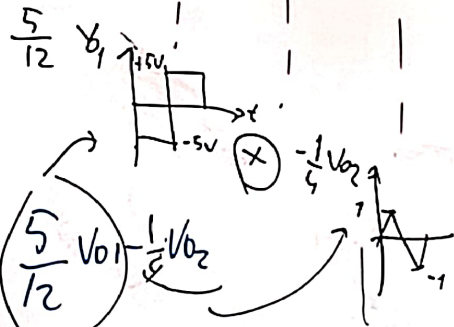
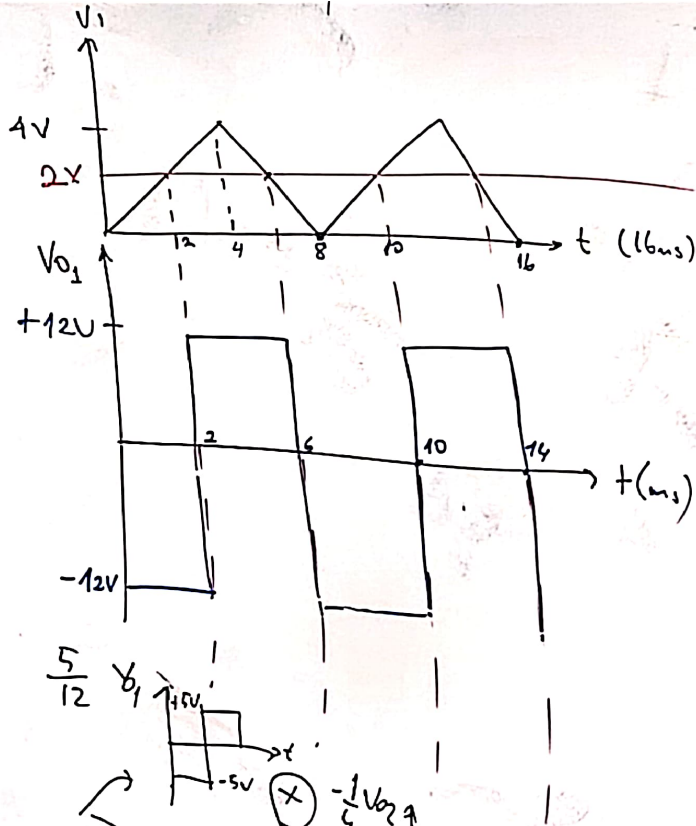
$$T = 16ms$$

V_{O1}, V_{O2}, V_O 'ı özetli olarak çiz.

$$\frac{V_{O2} - V_{O1}/3}{4K} = \frac{V_{O1}/3 - V_O}{1K}$$

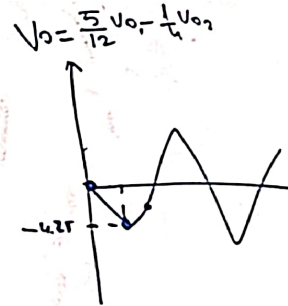
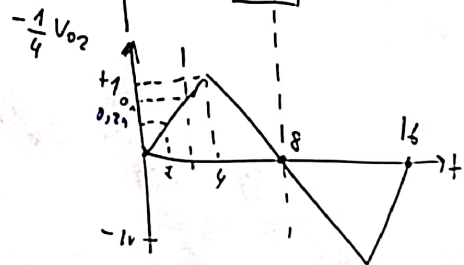
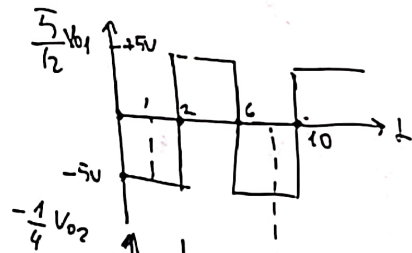
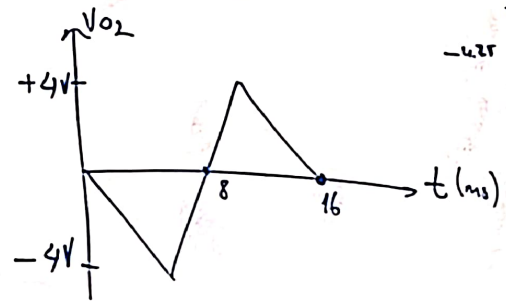
$$V_{O2} - \frac{V_{O1}}{3} = \frac{4}{3}V_{O1} - 4V_O$$

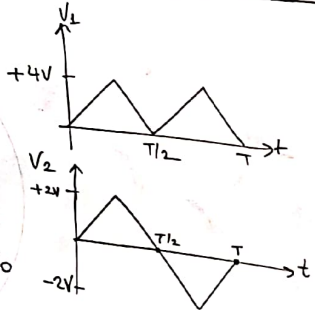
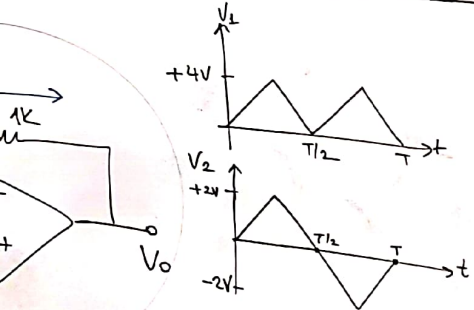
$$4V_O = \frac{5}{3}V_{O1} - V_{O2} \Rightarrow V_O = \frac{1}{4} \left(\frac{5}{3}V_{O1} - V_{O2} \right) = \frac{5}{12}V_{O1} - \frac{1}{4}V_{O2}$$



$$V_{O2} = -V_2 \left(\frac{10K}{5K} \right)$$

$$V_{O2} = -2V_2$$





$$\pm V_{cc} = \pm 12V$$

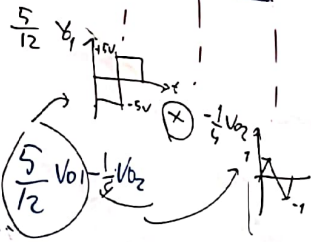
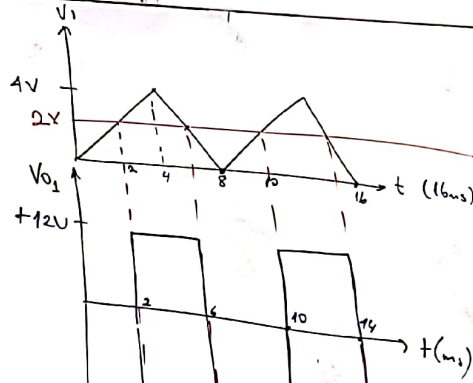
$$T = 16ms$$

V_{01}, V_{02}, V_0 'ı özetli olarak çiz.

$$\frac{V_{02} - V_{01}/3}{4K} = \frac{V_{01}/3 - V_0}{1K}$$

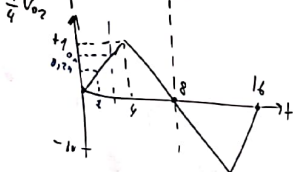
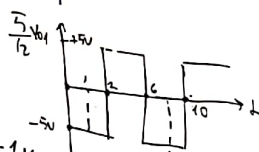
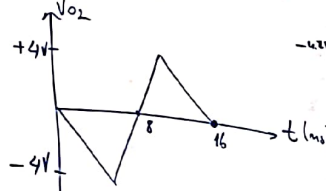
$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

$$4V_0 = \frac{5}{3}V_{01} - V_{02} \Rightarrow V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right) = \left(\frac{5}{12}V_{01} - \frac{1}{4}V_{02} \right)$$

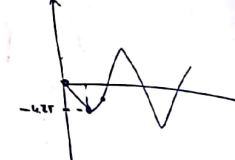


$$V_{02} = -V_2 \left(\frac{10K}{5K} \right)$$

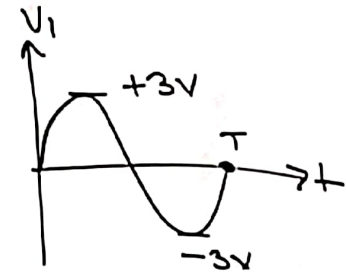
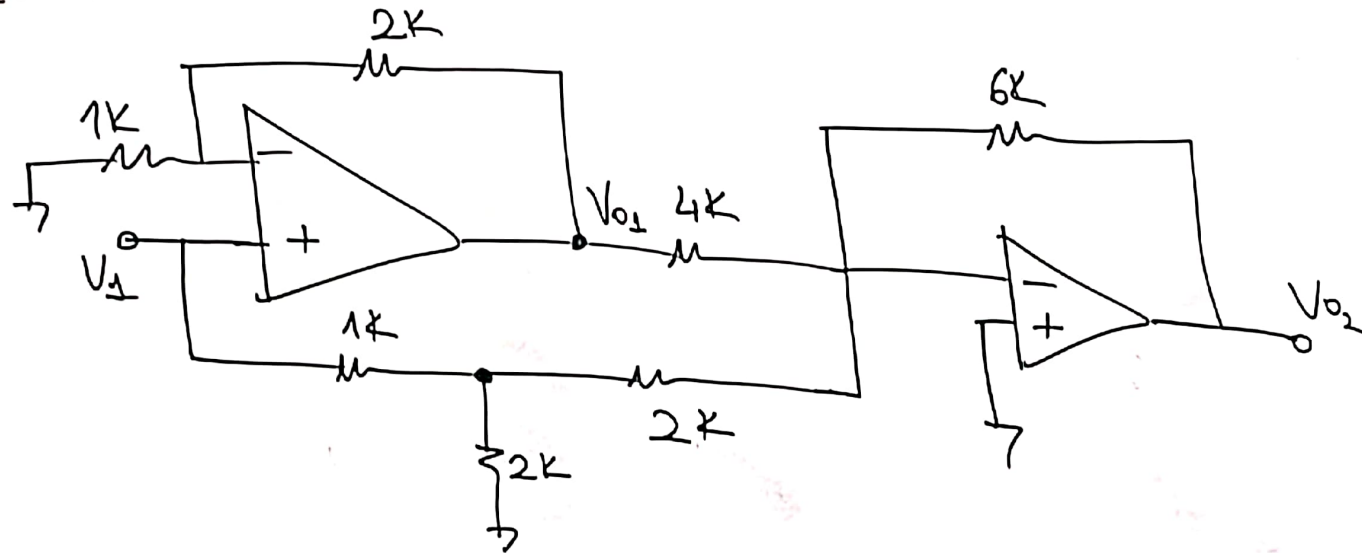
$$V_{02} = -2V_2$$



$$V_0 = \frac{5}{12}V_{01} - \frac{1}{4}V_{02}$$



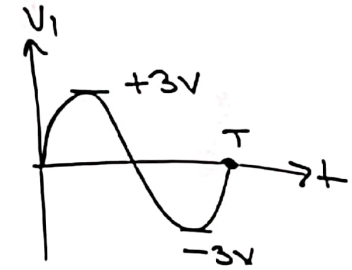
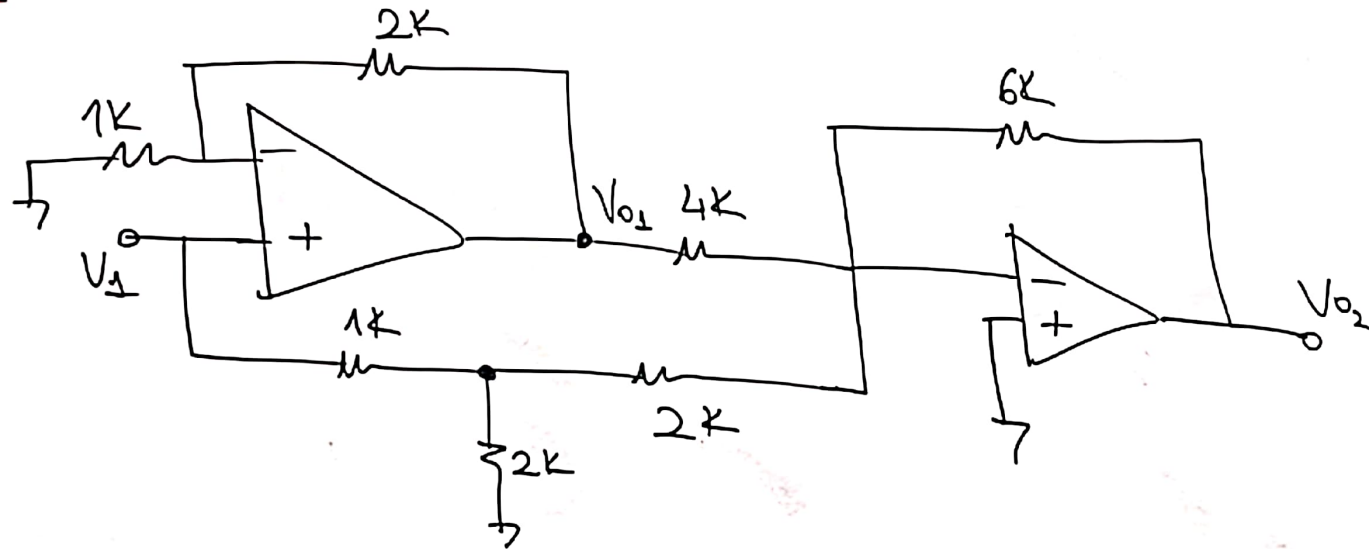
2024-Büt.



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

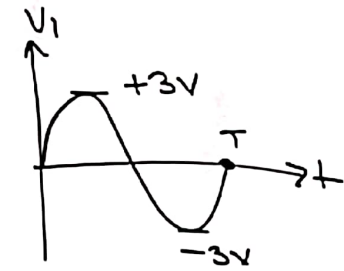
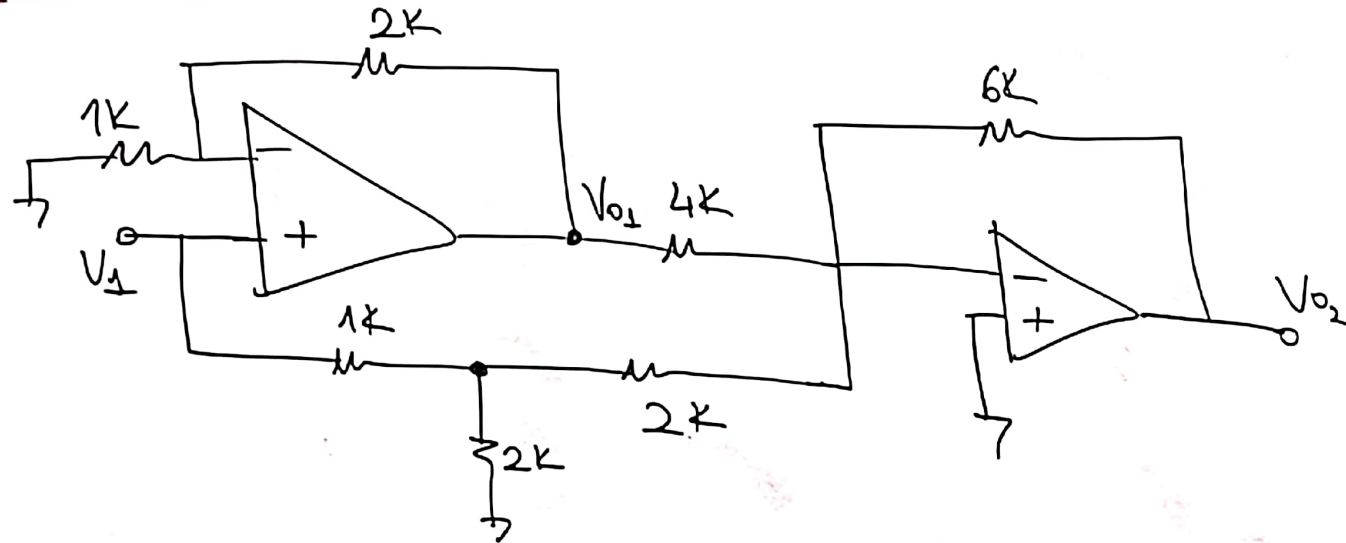
2024-Büt.



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

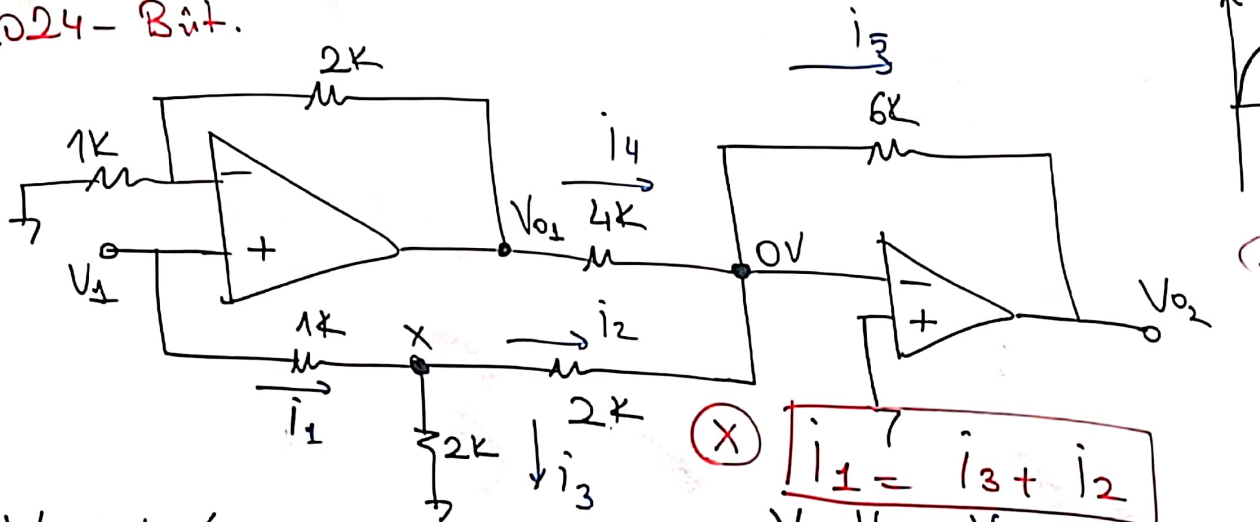
2024-Büt.



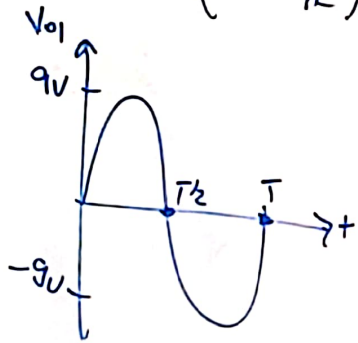
$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

2024 - Büt.



$$V_{01} = V_1 \cdot \left(1 + \frac{2k}{1k}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$



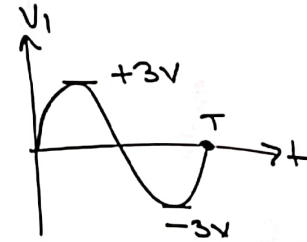
$$\textcircled{X} \quad \boxed{i_1 = i_3 + i_2}$$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x}$$

$$V_x = \frac{V_1}{2}$$



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} girsin.

$$i_2 + i_4 = i_5$$

$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

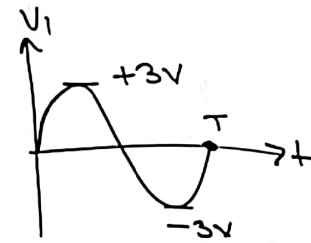
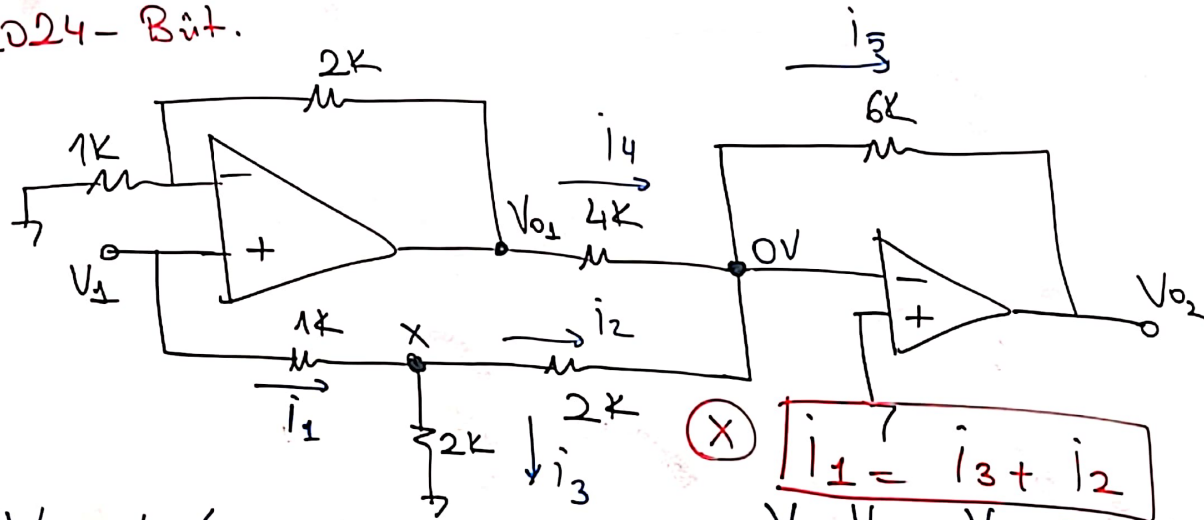
$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\boxed{V_{02} = -6V_1}$$



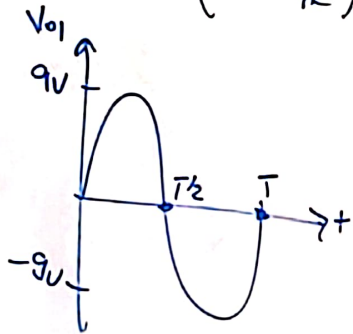
2024 - Büt.



$\pm V_{CC} = \pm 15V$

V_{01}, V_{02} gain.

$$V_{01} = V_1 \cdot \left(1 + \frac{2K}{1K}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$



(X) $\boxed{i_1 = i_3 + i_2}$

$$\frac{V_1 - V_x}{1K} = \frac{V_x - 0}{2K} + \frac{V_x - 0}{2K}$$

$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x}$$

$$V_x = \frac{V_1}{2}$$

$$i_2 + i_4 = i_5$$

$$\frac{V_{01} - 0}{4K} + \frac{V_x - 0}{2K} = \frac{0 - V_{02}}{6K}$$

$$\frac{3V_1}{4K} + \frac{V_1/2}{2K} = -\frac{V_{02}}{6K}$$

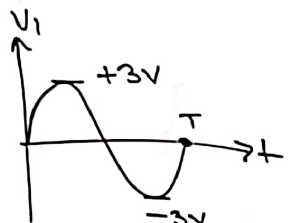
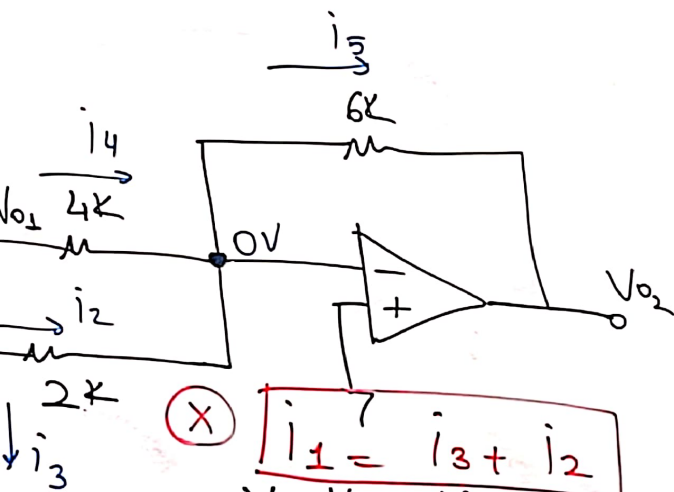
$$\frac{3V_1}{4K} + \frac{V_1}{4K} = -\frac{V_{02}}{6K}$$

$$\frac{4V_1}{4K} = -\frac{V_{02}}{6K}$$

$$\boxed{V_{02} = -6V_1}$$

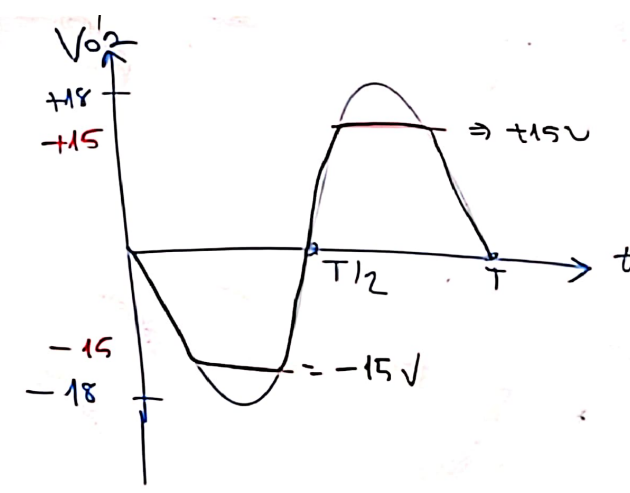
+18
+15

-15
-18



$\pm V_{CC} = \pm 15V$

V_{O1}, V_{O2} are in.



(X) $i_1 = i_3 + i_2$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

$$V_1 - V_x = V_x$$

$$V_1 = 2V_x$$

$$V_x = \frac{V_1}{2}$$

$$i_2 + i_4 = i_3$$

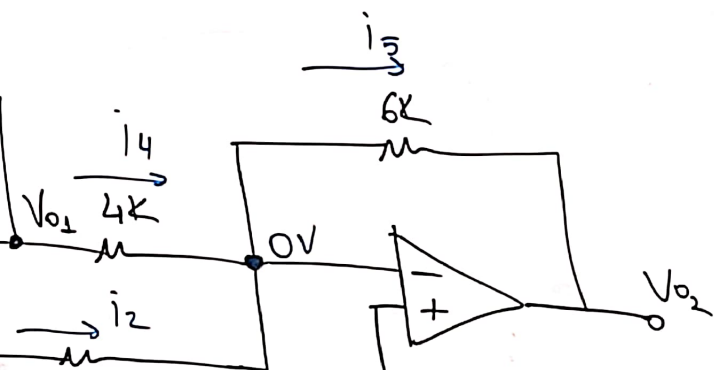
$$\frac{V_{O1} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{O2}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{O2}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{O2}}{6k}$$

$$\frac{4V_1}{4k} = -\frac{V_{O2}}{6k}$$

$$V_{O2} = -6V_1$$



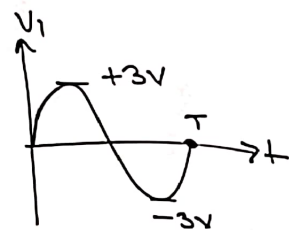
(X) $i_1 = i_3 + i_2$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

$$V_1 - V_x = V_x$$

$$V_1 = 2V_x$$

$$V_x = V_1/2$$



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

$$i_2 + i_4 = i_5$$

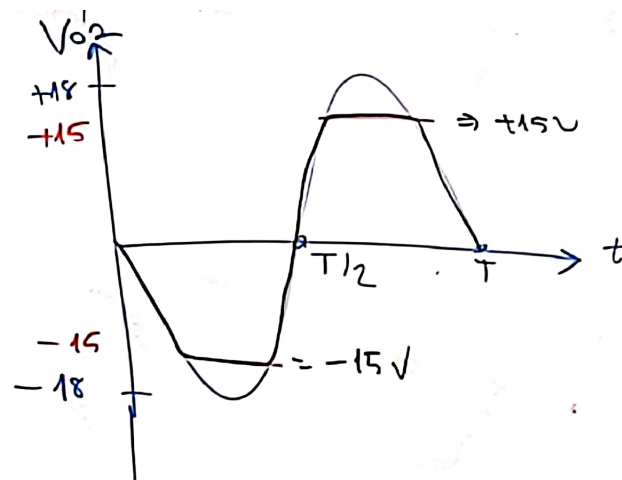
$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

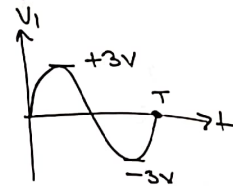
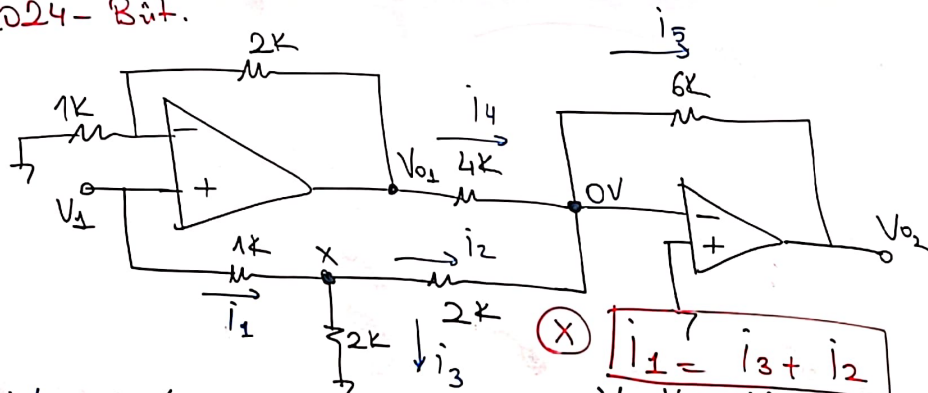
$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k}$$

$$V_{02} = -6V_1$$



2024 - Bit.



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

$$i_2 + i_4 = i_5$$

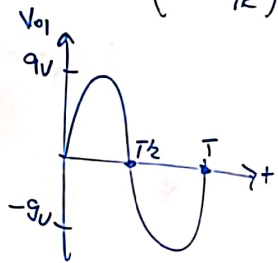
$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k} \quad \boxed{V_{02} = -6V_1}$$

$$V_{01} = V_1 \cdot \left(1 + \frac{2k}{1k}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$



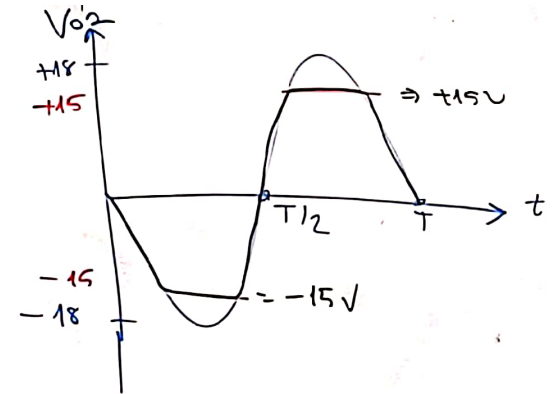
$$\textcircled{X} \quad i_1 = i_3 + i_2$$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

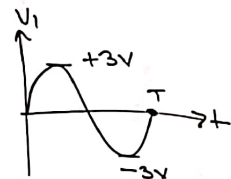
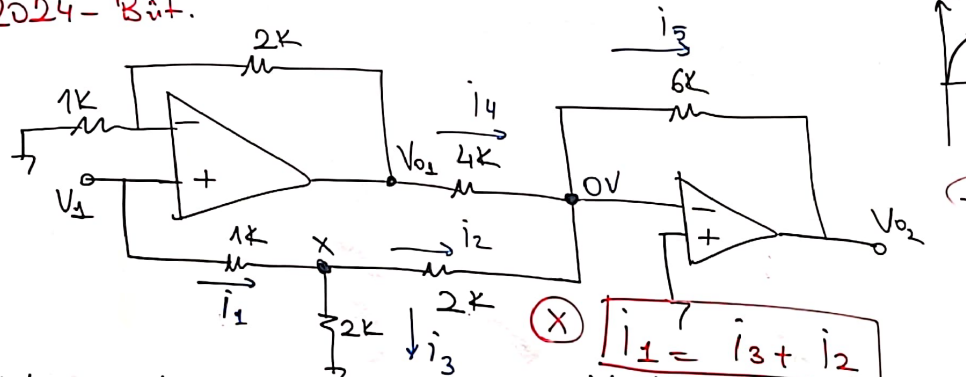
$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x}$$

$$V_x = \frac{V_1}{2}$$

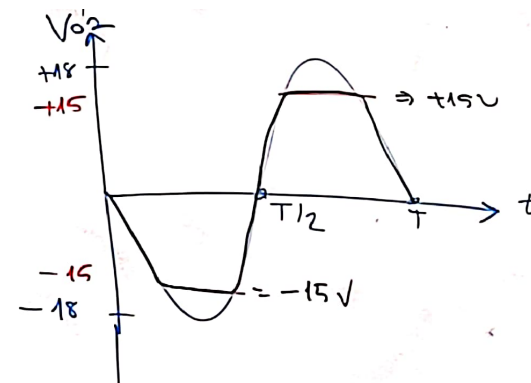


2024 - Büt.

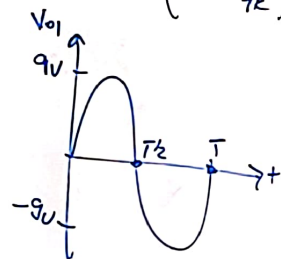


$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.



$$V_{01} = V_1 \cdot \left(1 + \frac{2K}{1K}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$



$$\textcircled{X} \quad \boxed{i_1 = i_3 + i_2}$$

$$\frac{V_1 - V_x}{1K} = \frac{V_x - 0}{2K} + \frac{V_x - 0}{2K}$$

$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x}$$

$$V_x = \frac{V_1}{2}$$

$$i_2 + i_4 = i_5$$

$$\frac{V_{01} - 0}{4K} + \frac{V_x - 0}{2K} = \frac{0 - V_{02}}{6K}$$

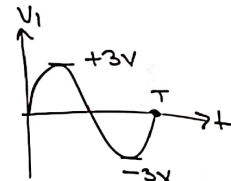
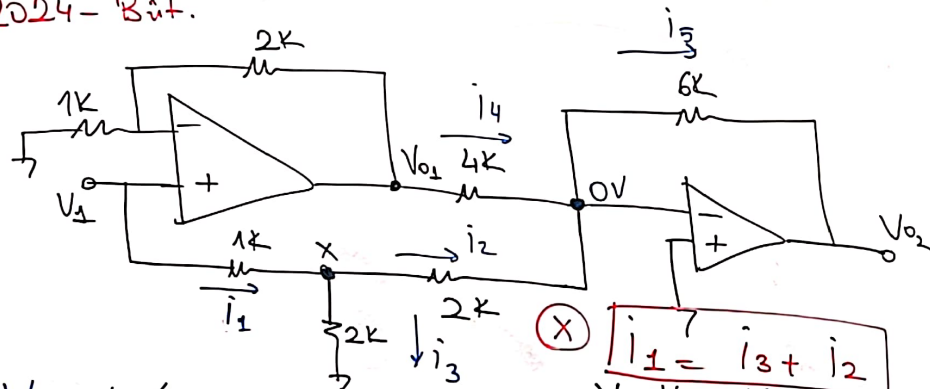
$$\frac{3V_1}{4K} + \frac{V_1/2}{2K} = -\frac{V_{02}}{6K}$$

$$\frac{3V_1}{4K} + \frac{V_1}{4K} = -\frac{V_{02}}{6K}$$

$$\frac{4V_1}{4K} = -\frac{V_{02}}{6K}$$

$$\boxed{V_{02} = -6V_1}$$

2024-Bit.



$$\pm V_{CC} = \pm 15V$$

V_{01}, V_{02} gain.

$$i_2 + i_4 = i_5$$

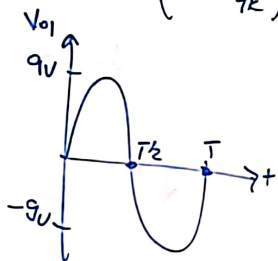
$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k} \Rightarrow \boxed{V_{02} = -6V_1}$$

$$V_{01} = V_1 \cdot \left(1 + \frac{2k}{1k}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$



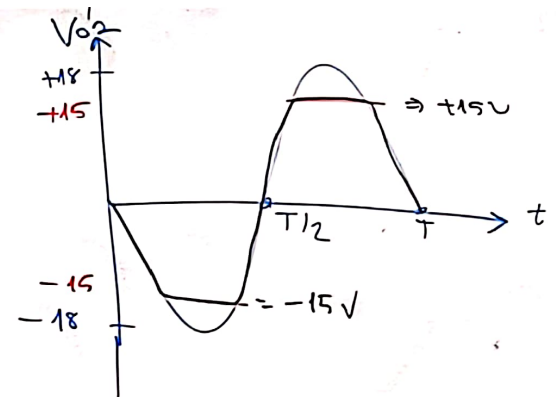
$$\boxed{i_1 = i_3 + i_2}$$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

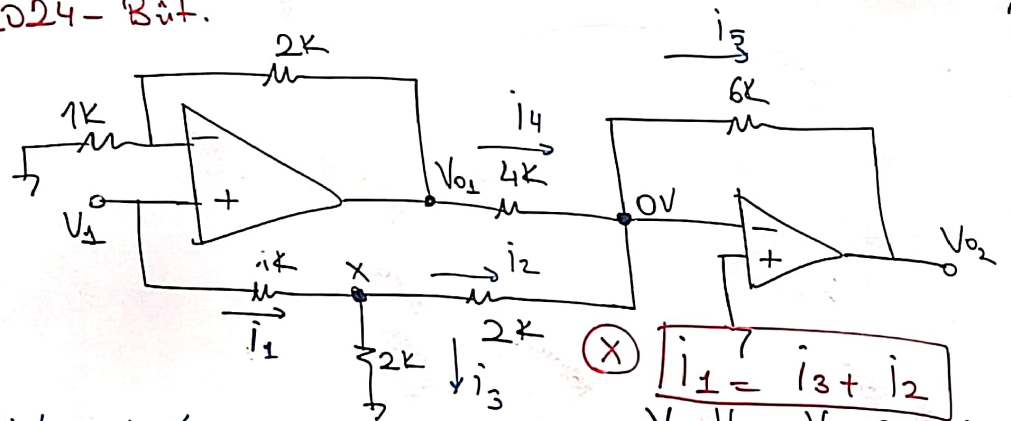
$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x}$$

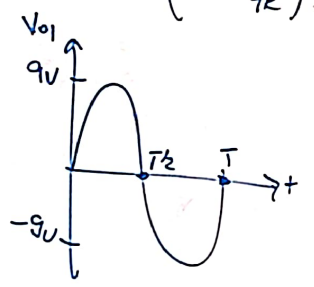
$$V_x = \frac{V_1}{2}$$



2024 - Büt.



$$V_{01} = V_1 \cdot \left(1 + \frac{2k}{1k}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$

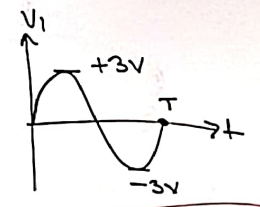


$$\textcircled{X} \quad \boxed{i_1 = i_3 + i_2}$$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x} \quad V_x = \frac{V_1}{2}$$



$$\pm V_{cc} = \pm 15V$$

V_{01}, V_{02} gain.

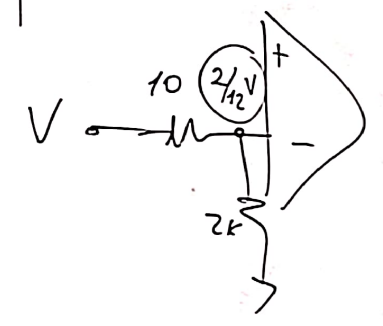
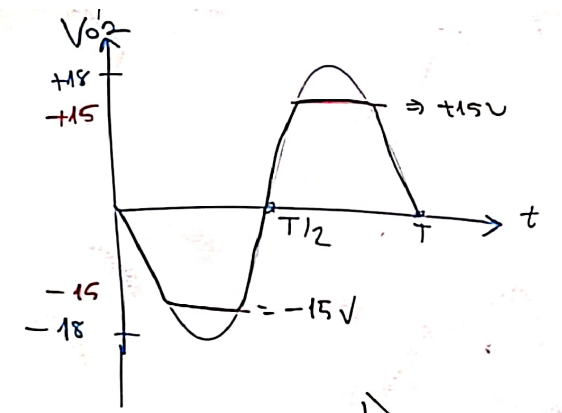
$$i_2 + i_4 = i_5$$

$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

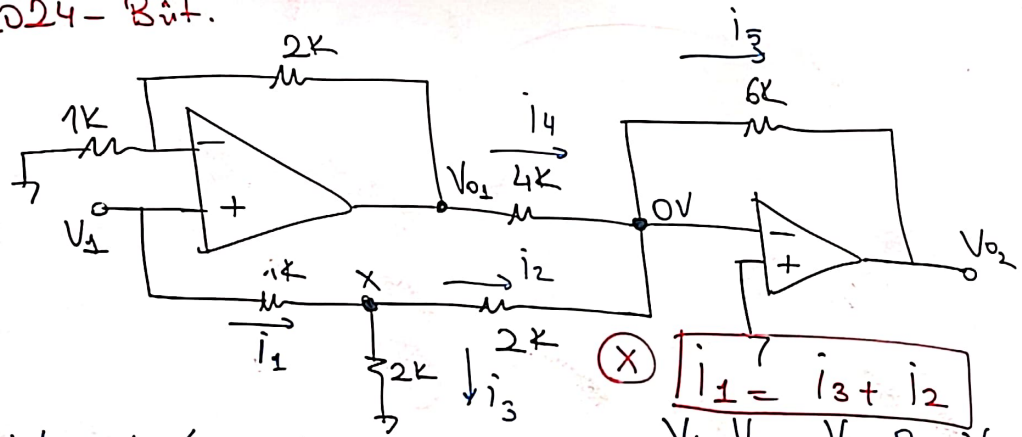
$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

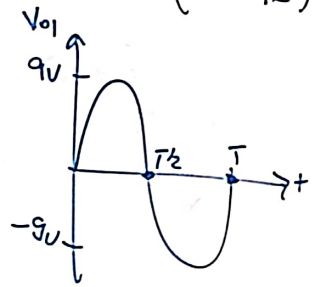
$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k} \quad \boxed{V_{02} = -6V_1}$$



2024-Büt.



$$V_{01} = V_1 \cdot \left(1 + \frac{2k}{1k}\right) \Rightarrow \boxed{V_{01} = 3V_1}$$

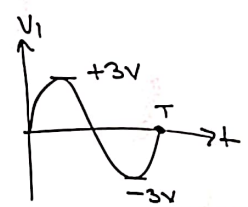


(X) $i_1 = i_3 + i_2$

$$\frac{V_1 - V_x}{1k} = \frac{V_x - 0}{2k} + \frac{V_x - 0}{2k}$$

$$V_1 - V_x = V_x$$

$$\boxed{V_1 = 2V_x} \quad V_x = \frac{V_{01}}{2}$$



$\pm V_{CC} = \pm 15V$
 V_{01}, V_{02} girsin.

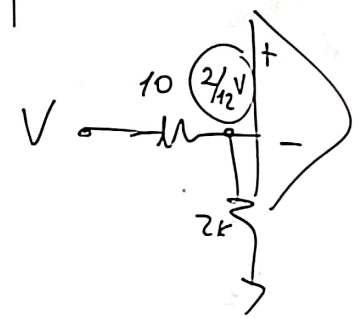
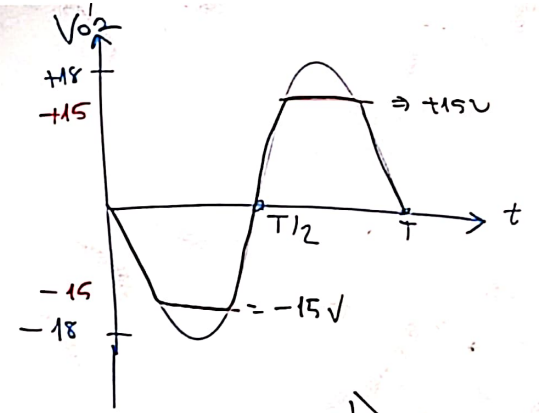
$$i_2 + i_4 = i_5$$

$$\frac{V_{01} - 0}{4k} + \frac{V_x - 0}{2k} = \frac{0 - V_{02}}{6k}$$

$$\frac{3V_1}{4k} + \frac{V_1/2}{2k} = -\frac{V_{02}}{6k}$$

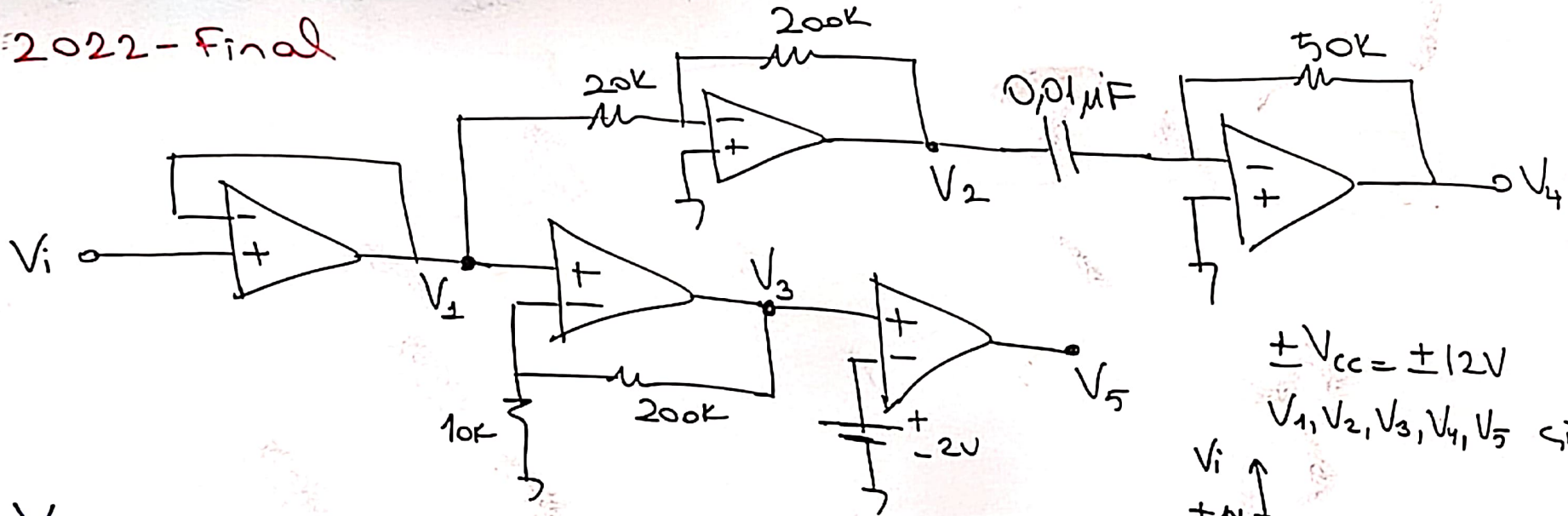
$$\frac{3V_1}{4k} + \frac{V_1}{4k} = -\frac{V_{02}}{6k}$$

$$\frac{4V_1}{4k} = -\frac{V_{02}}{6k} \Rightarrow \boxed{V_{02} = -6V_1}$$



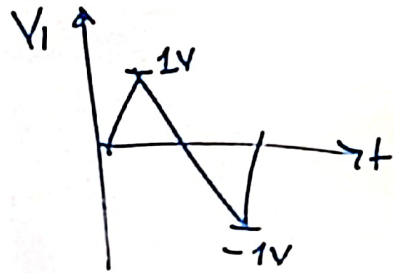
Akim tek bir cümeden ikenliyorsu
geniim bōlōō yapabiliñiz.

2022-Final



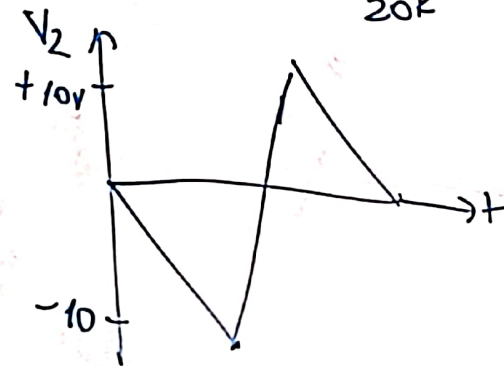
V_1 , voltaj izleyici

$$V_1 = V_i$$



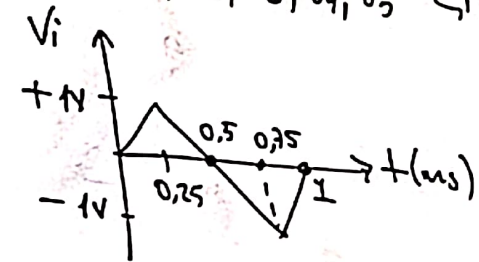
V_2 , faz çeviren kuv.

$$V_2 = -V_1 \cdot \frac{200K}{20K} = -10V_1$$

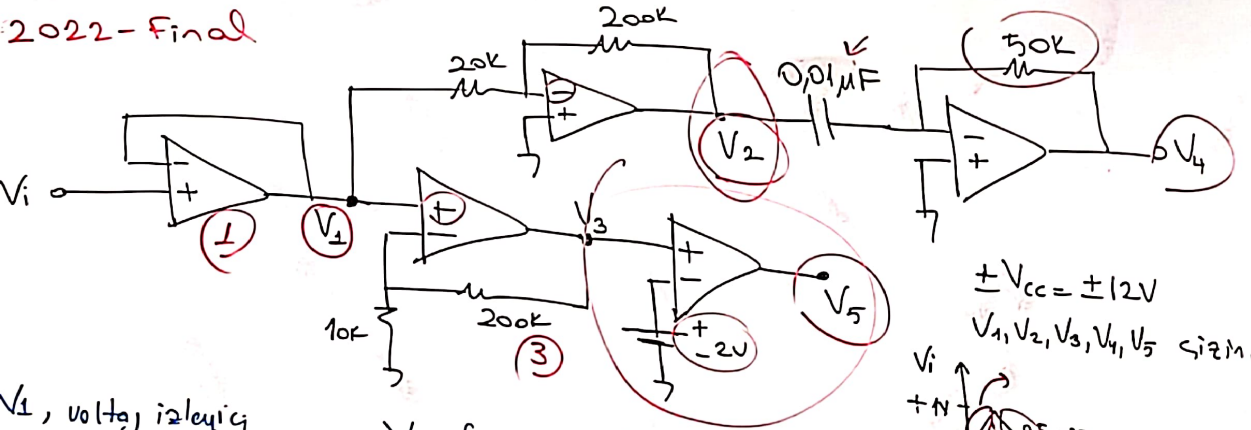


$$\pm V_{CC} = \pm 12V$$

V_1, V_2, V_3, V_4, V_5 çizim.

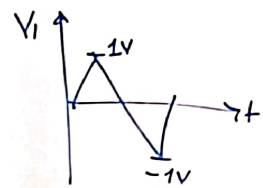


2022-Final



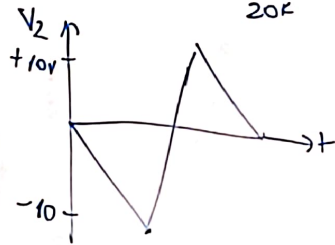
V_1 , voltaj izleyici

$$V_1 = V_i$$



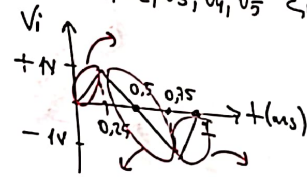
V_2 , faz çeviren kuv.

$$V_2 = -V_1 \cdot \frac{200k}{20k} = -10V_1$$



$$\pm V_{CC} = \pm 12V$$

V_1, V_2, V_3, V_4, V_5 çizim.

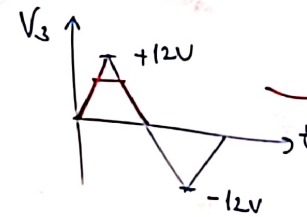


V_3 , faz çevirmeyen kuv.

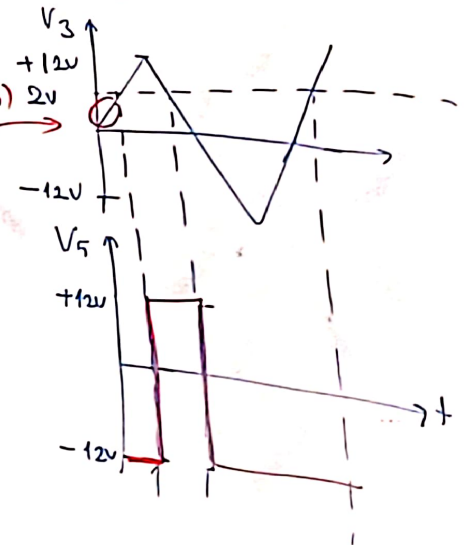
$$V_3 = \left(1 + \frac{200k}{10k}\right) \cdot V_1$$

$$V_3 = 21V_1$$

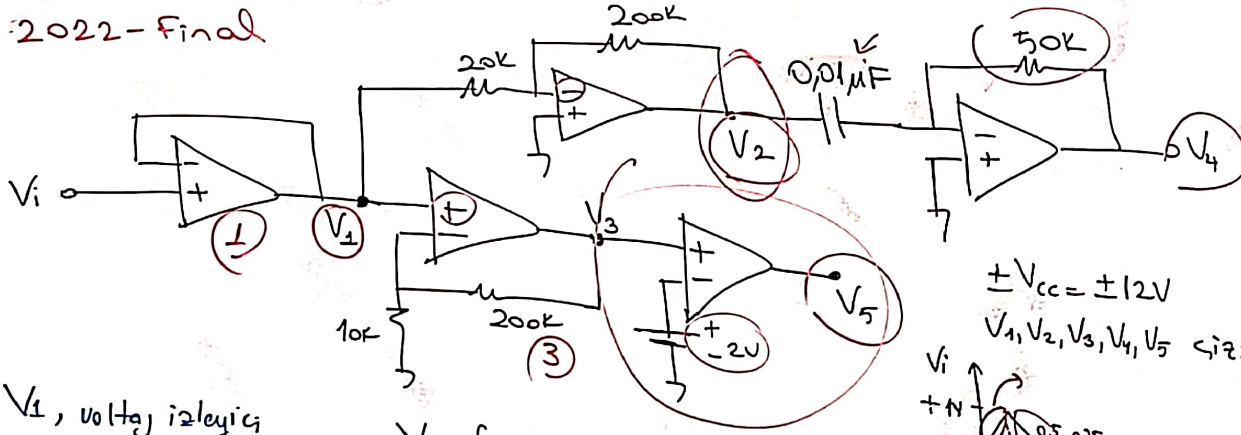
$\pm V_{CC} = \pm 12V$ old. göre



V_5 , karşılaştırıcı

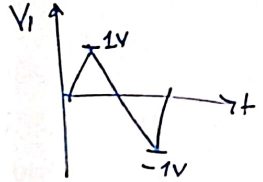


2022-Final



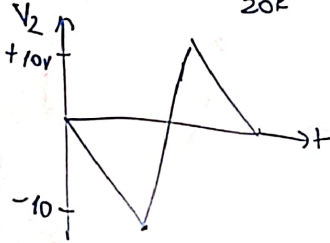
V_1 , voltaj izleyici

$$V_1 = V_i$$



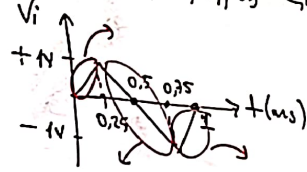
V_2 , faz çeviren kuv.

$$V_2 = -V_1 \cdot \frac{200k}{20k} = -10V_1$$



$$\pm V_{cc} = \pm 12V$$

V_1, V_2, V_3, V_4, V_5 çizim.

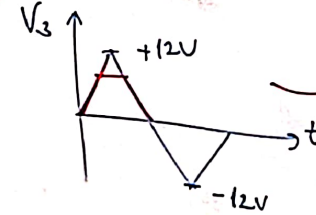


V_3 , faz çevirmeyen kuv.

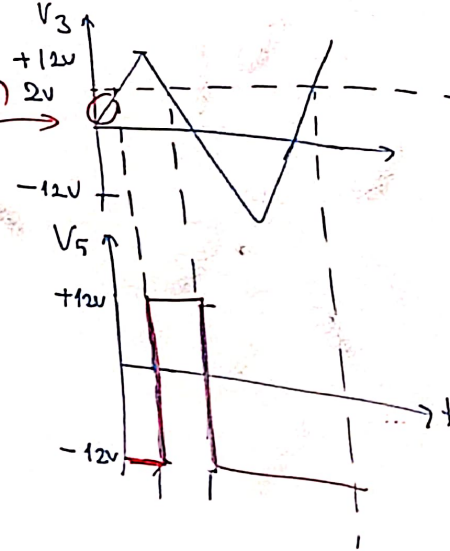
$$V_3 = \left(1 + \frac{200k}{10k}\right) \cdot V_1$$

$$V_3 = 21V_1$$

$\pm V_{cc} = \pm 12V$, old. göre



V_5 , karşılaştırıcı

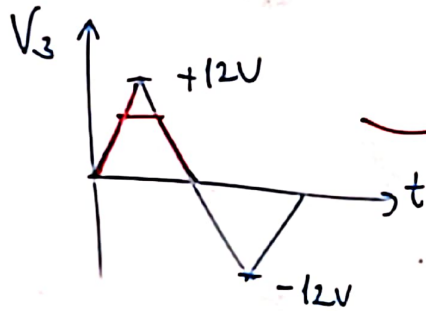


V_3 , faz çevirmeyen kuv.

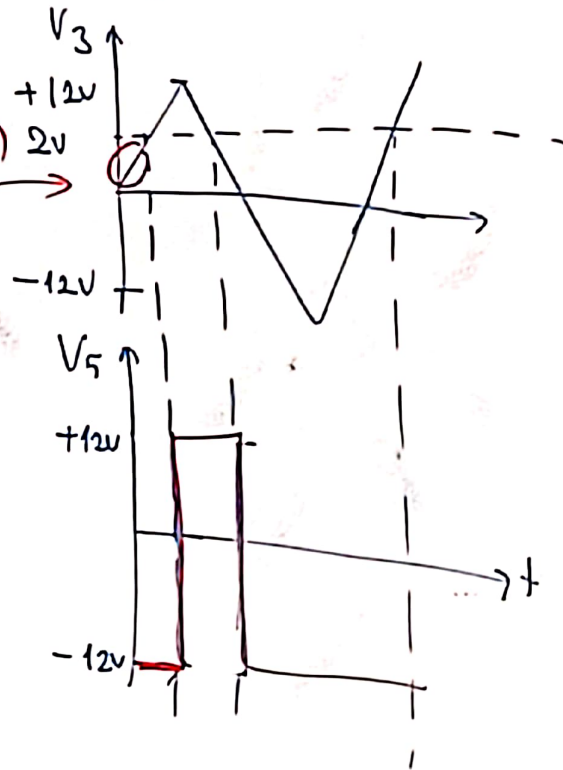
$$V_3 = \left(1 + \frac{200K}{10K}\right) \cdot V_1$$

$$V_3 = 21 V_1$$

$\pm V_{CC} = \pm 12V$ old. göre



V_5 , karşılaştırmacı



V_4 , türev alma,

$$V_4(t) = -R.C. \frac{dV_2(t)}{dt} = -(50 \cdot 10^3) \cdot \left(40 \cdot 10^{-3}\right) \cdot \frac{dV_2}{dt}$$

$$0 < t < 0,25ms \Rightarrow V_2(t) = -40 \cdot 10^3 t$$

$$V_4(t) = +20$$

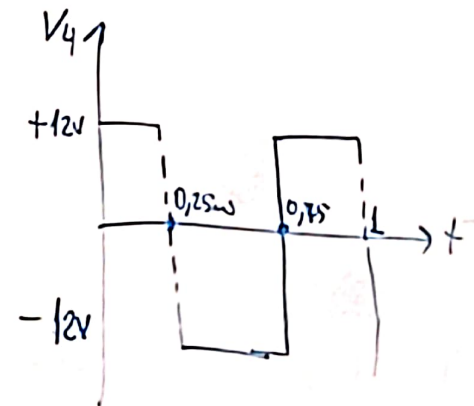
$$0,25ms < t < 0,75ms \Rightarrow V_2(t) = 40 \cdot 10^3 t - 20$$

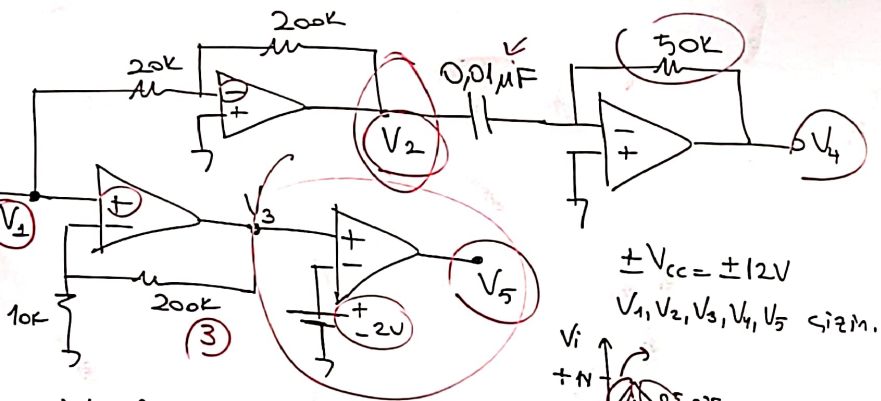
$$V_4(t) = -20$$

$$0,75ms < t < 1ms \Rightarrow V_2(t) = -40 \cdot 10^3 t + 40$$

$$V_4(t) = +20$$

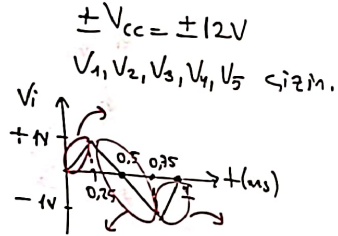
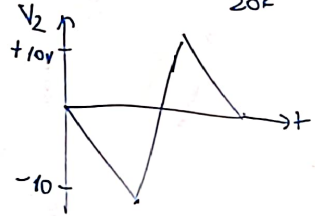
$\pm V_{CC} = \pm 12V$ old. için





V_2 , faz çeviren kuv.

$$V_2 = -V_1 \cdot \frac{200K}{20K} = -10V_1$$

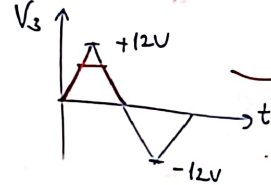


V_3 , faz çevirmeyen kuv.

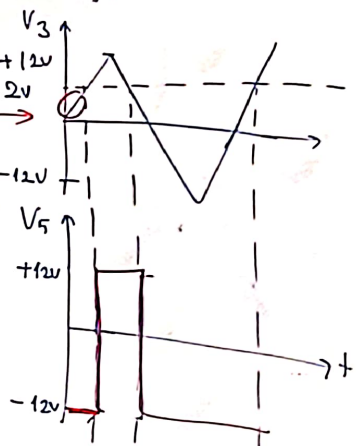
$$V_3 = \left(1 + \frac{200K}{10K}\right) \cdot V_1$$

$$V_3 = 21V_1$$

$\pm V_{CC} = \pm 12V$, old. göre



V_5 , karşılaştırıcı



V_4 , türev alma,

$$V_4(t) = -R.C \cdot \frac{dV_2(t)}{dt} = -(50 \cdot 10^3) \cdot (0.01 \cdot 10^{-6}) \cdot \frac{dV_2}{dt}$$

$$0 < t < 0.25ms \Rightarrow V_2(t) = -40 \cdot 10^3 t$$

$$V_4(t) = +20$$

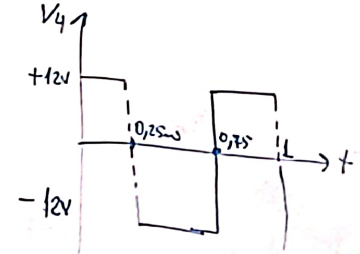
$$0.25ms < t < 0.75ms \Rightarrow V_2(t) = 40 \cdot 10^3 t - 20$$

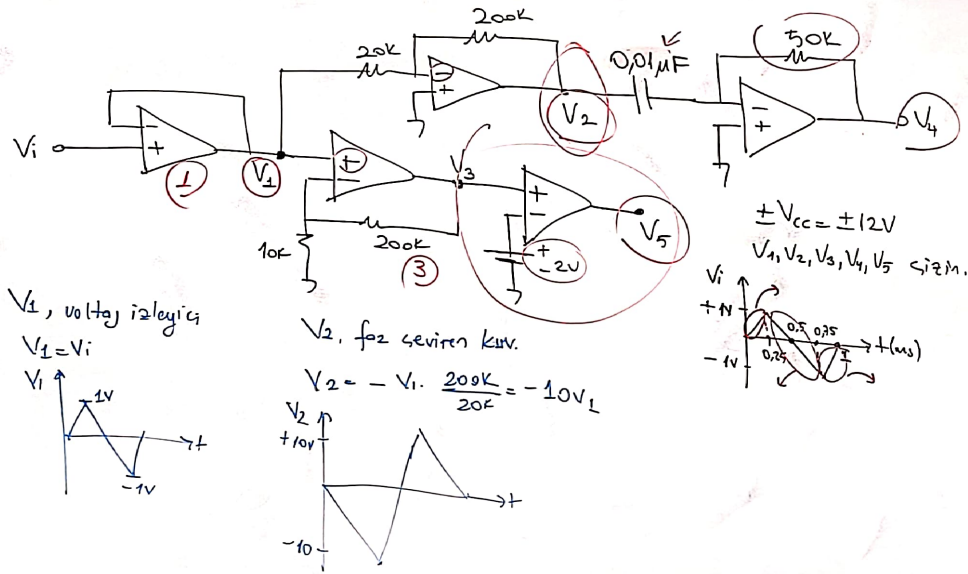
$$V_4(t) = -20$$

$$0.75ms < t < 1ms \Rightarrow V_2(t) = -40 \cdot 10^3 t + 40$$

$$V_4(t) = +20$$

$\pm V_{CC} = \pm 12V$ old. için



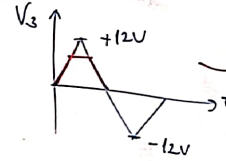


V_3 , faz çevirmeyen kuv.

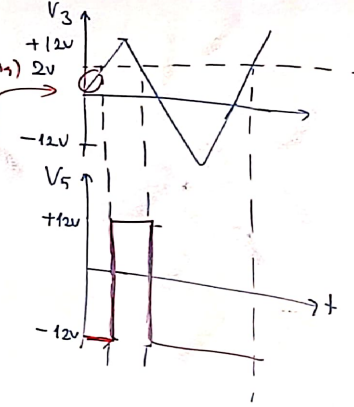
$$V_3 = \left(1 + \frac{200k}{10k}\right) \cdot V_2$$

$$V_3 = 21V_2$$

$$\pm V_{cc} = \pm 12V \text{ old. göre}$$



V_5 , karşılaştırıcı



V_4 , türev alma

$$V_4(t) = -R.C. \frac{dV_3(t)}{dt} = -(50k)(0.01\mu F) \cdot \frac{dV_3}{dt}$$

$$0 < t < 0.25ms \Rightarrow V_2(t) = -40 \cdot 10^3 t$$

$$V_4(t) = +20$$

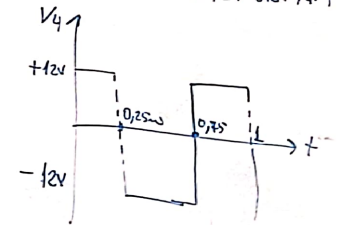
$$0.25ms < t < 0.75ms \Rightarrow V_2(t) = 40 \cdot 10^3 t - 20$$

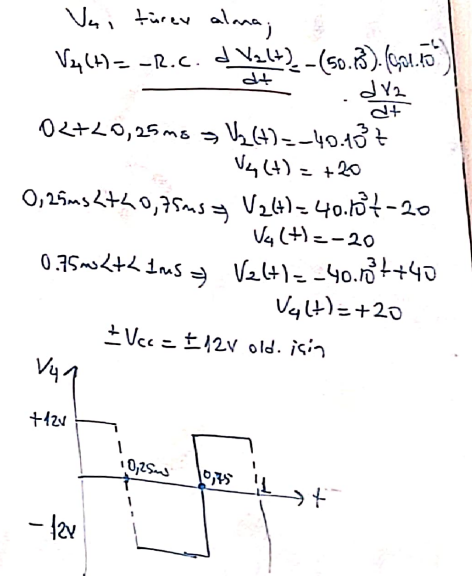
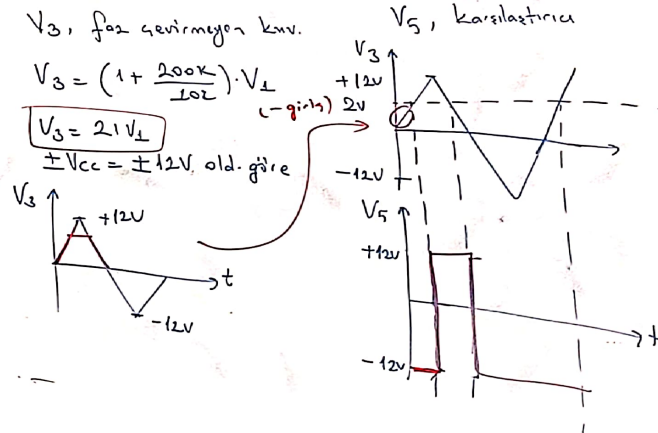
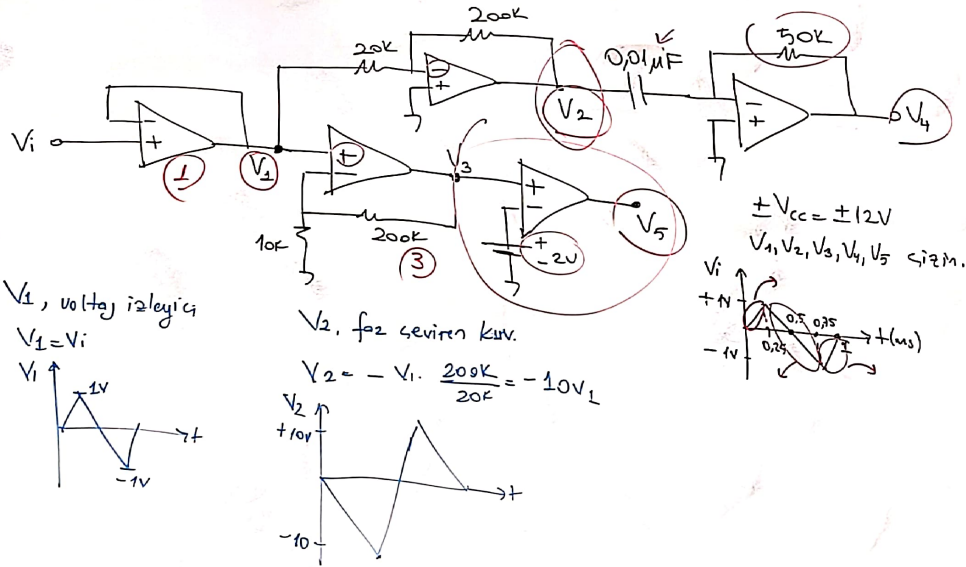
$$V_4(t) = -20$$

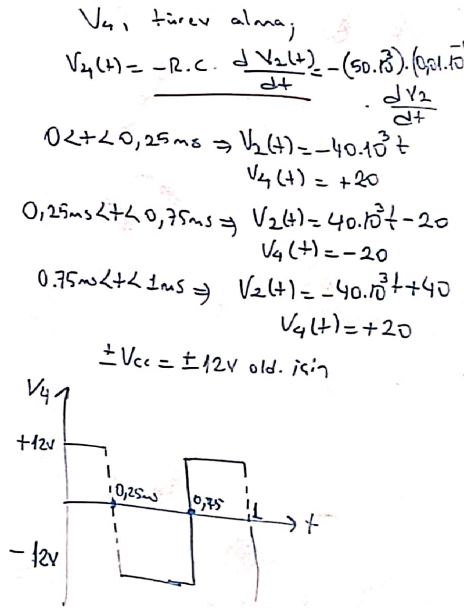
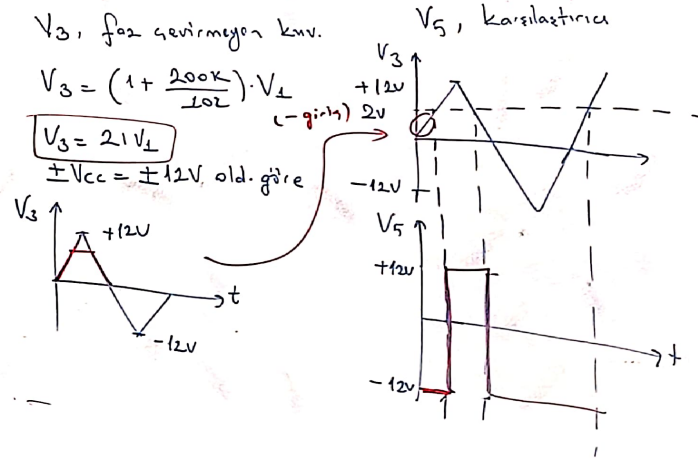
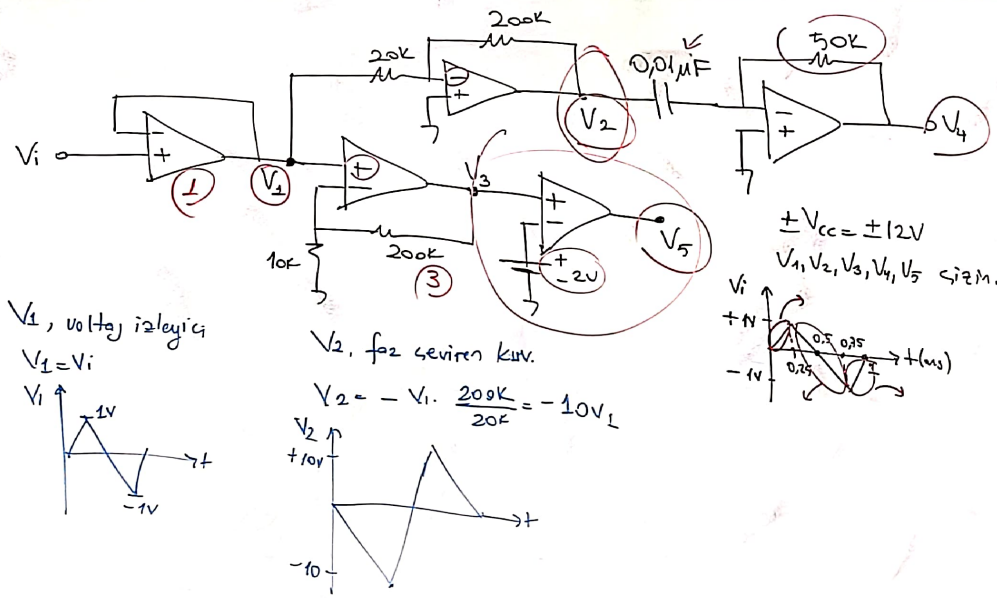
$$0.75ms < t < 1ms \Rightarrow V_2(t) = -40 \cdot 10^3 t + 40$$

$$V_4(t) = +20$$

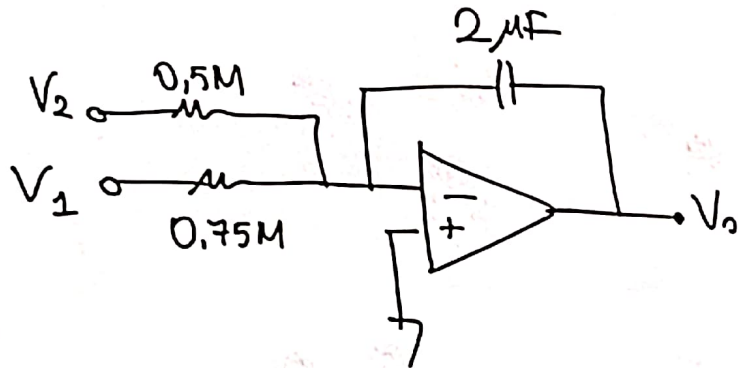
$\pm V_{cc} = \pm 12V$ old. için





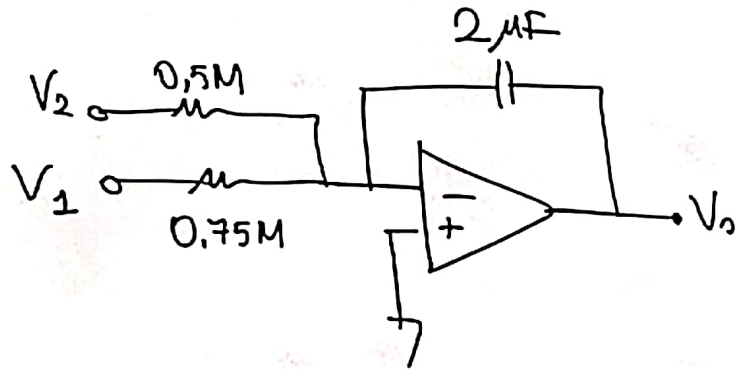


Final Sorusu:



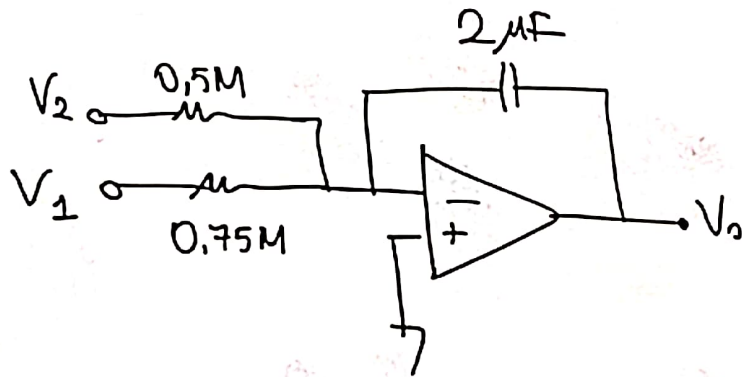
$$\left. \begin{array}{l} V_1 = 10 \sin(200t) \\ V_2 = 15 \sin(200t) \\ \pm V_{cc} = \pm 15V \end{array} \right\} \Rightarrow V_0 \text{ 'ı cizil.} \\ (V_0(t) \text{ 'yi elde ediniz})$$

Final Sorusu:



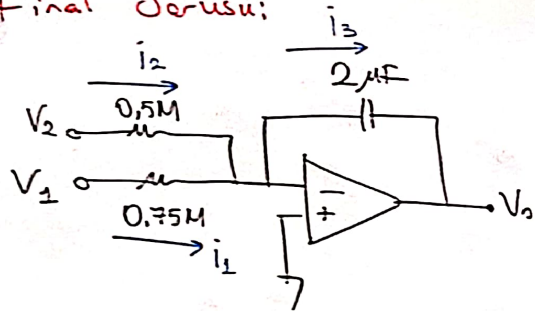
$$\left. \begin{array}{l} V_1 = 10 \sin(200t) \\ V_2 = 15 \sin(200t) \\ \pm V_{cc} = \pm 15V \end{array} \right\} \Rightarrow V_o \text{ 'ı gizin.} \\ (V_o(t) \text{ 'y' elde ediniz})$$

Final Sorusu:



$$\left. \begin{array}{l} V_1 = 10 \sin(200t) \\ V_2 = 15 \sin(200t) \\ \pm V_{cc} = \pm 15V \end{array} \right\} \Rightarrow V_o \text{ 'ı cizil.} \\ (V_o(t) \text{ 'yi elde ediniz})$$

Final Sorusu:



$$\left. \begin{aligned} V_1 &= 10 \sin(200t) \\ V_2 &= 15 \sin(200t) \\ \pm V_{cc} &= \pm 15V \end{aligned} \right\} \Rightarrow V_{0.1} \text{ çizil.} \\ (V_0(t) \text{ 'u' elde ediniz})$$

$$i_1 + i_2 = i_3$$

$$\frac{V_2 - 0}{R_1} + \frac{V_2 - 0}{R_2} = \frac{0 - V_0}{X_c}$$

$$\boxed{\text{NOT: } \sin(200t) = \cos(200t - 90^\circ) = 1 \angle 90^\circ = -j}$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} = \frac{-V_0}{1/j\omega C}$$

$$V_0 = -\frac{V_2}{j\omega C R_2} - \frac{V_1}{j\omega C R_1}$$

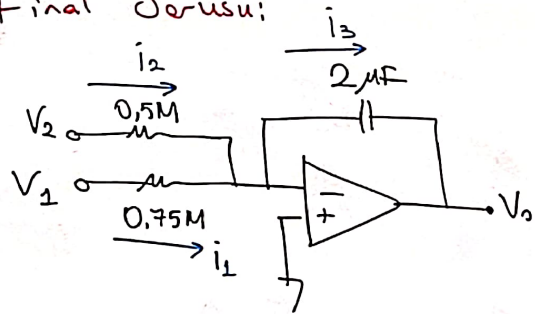
$$V_0 = -\frac{-j15}{j(200)(2 \cdot 10^{-6})(0.5 \cdot 10^6)} - \frac{-j10}{j(200)(2 \cdot 10^{-6})(0.75 \cdot 10^6)}$$

$$V_0 = \frac{3}{40} + \frac{1}{30}$$

$V_0(t)$ ise;

$$V_0(t) = \left(\frac{3}{40} + \frac{1}{30} \right) \cos(200t)$$

Final Sorusu:



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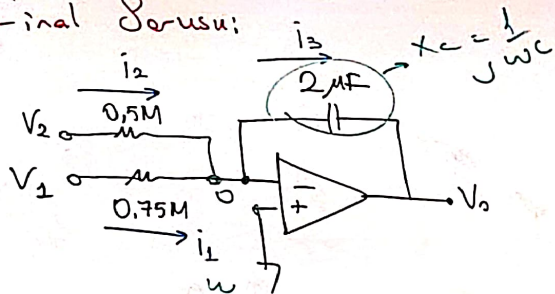
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